Discover Internet of Things

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Systematic review on securing IoT systems with post quantum cryptography emerging threats, countermeasures, and future research needs

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Abstract

The rapid deployment of Internet of Things (IoT) devices across critical infrastructure's sectors such as healthcare, transportation, and smart cities brings unprecedented security challenges; particularly as quantum computing threatens the classical cryptography. This systematic review analyzes IoT security in relation to post-quantum cryptography (PQC), highlighting vulnerabilities, evolving threat, and cryptographic countermeasures. Using the PRISMA 2020 protocol, our work employed a rigorous and reproducible search strategy, where the initial Boolean query, (("IoT" OR "Internet of Things") AND ("Cryptography" OR "Encryption Scheme") AND "Quantum"), was developed and effectively customized for nine reputable databases to maximize the coverage. From an initial pool of 5,151 publications dated between 2020 and February 12, 2025, a content-based screening was conducted at the title, abstract, and full-text levels. This meticulous filtering resulted in a final corpus of 161 studies deemed relevant for in-depth analysis and synthesis. Then, we categorize and analyze PQC algorithms by vulnerabilities, suitability for resource-constrained IoT environments, performance evaluation, scalability, and quantum resilience. We also identify key limitations in implementations of current cryptographic schemes, including issues in key management and authentication; and outline major challenges such as resource constraints of IoT devices, environmental noise, quantum reversible computing, security vulnerabilities, quantum hardware constraints, and complexities in integration. To counter these threats, we propose a framework of PQC countermeasures and illustrated by a hypothetical case study of PQC adoption in a real-world IoT system. Finally, we propose future research directions in lightweight PQC, hybrid frameworks, improved key management and distribution, addressing quantum distance matters, countering quantum threats, and standardization for IoT environments.

Article Highlights

- Identifies the cryptographic weaknesses in current IoT systems that are vulnerable to quantum attacks and outlines the resulting threat landscape.

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- Evaluates leading post-quantum cryptography families to identify the most practical solutions for resource-constrained IoT devices.
- Outlines a strategic roadmap for implementation, highlighting challenges and a path to future-resilient IoT security.

Keywords Resource-Constrained devices, Securing IoT systems, Post-Quantum cryptography, Quantum emerging threats, Cryptographic countermeasures, Hybrid cryptographic framework, Lightweight cryptography, Systematic review

1 Introduction

1.1 Background of the study

The advancements of using IoT devices in various sectors are transforming and simplifying the way the industries operate and provide services. So far billions of IoT devices are interconnected, and their number is steadily growing every year. These interconnected small devices generate and exchange a massive amount of sensitive data [1]. To protect this massive and sensitive data, robust cryptographic protocols are essential for ensuring its integrity, confidentiality, and availability. To achieve this, traditional cryptographic schemes, such as Rivest-Shamir-Adleman (RSA) and Elliptic Curve Cryptography (ECC) that rely on mathematical problems (e.g., prime factorization and discrete logarithms) are currently in use [2]. However, these schemes were built for classical computing environment and are increasingly getting challenged by the technological advancement and the growing landscape of everyday security threats. As a result, critical infrastructure, namely government intelligence, medical records, intellectual property, and financial data faced unprecedented exposure.

IoT devices are known to be resource-constrained with limitations in processing capability, memory size, and energy capacity. These inherent limitations necessitate the adoption of cryptographic protocols that are both secure and efficient in resource utilization. Consequently, lightweight encryption algorithms have emerged as a solution [3]. In addition to data encryption, these algorithms have also important roles in security services such as device authentication, key distribution, and secure bootstrapping, which are highly important in IoT communication [4]. Furthermore, resource-constrained IoT devices pose additional challenges for implementing classical cryptographic schemes, lightweight cryptography alternatives that often trade off security for efficiency. The threat of quantum computing thus exposes a critical gap: classical cryptographic schemes are increasingly at risk, and lightweight cryptographic solutions are not inherently quantum-resistant [5].

The introduction of quantum computers has consequently imposed additional threats to the existing traditional cryptographic protocols. Quantum computers leverage qubits and principles including superposition and entanglement as the powerful computational capability in order to easily solve the mathematical difficulties that are used as the basis for classical encryption algorithms. The quantum threat landscape is defined by foundational algorithmic threats: Shor's algorithm, which breaks the public-key cryptography underpinning the digital trust and secure connections, and Grover's algorithm, which weakens symmetric encryption. As such, it cascades into systemic risks like the "Harvest Now, Decrypt Later" (HNDL) attack by jeopardizing the long-term confidentiality of all data secured by current standards [6]. Such potential threats underscore the urgency of adapting cryptographic protocols to align with technological advancement

and safeguard IoT infrastructures against a wide range of attacks. In this context, the IoT domain faces unique and amplified risks due to its scale, resource constraints, and device lifespans. Though there is a growing research interest in post-quantum cryptographic solutions for securing IoT communications and data integrity, there is a critical gap still in obtaining a recent and comprehensive review that assesses the existing vulnerabilities, trending solutions, and available strategies for future research perspectives. The failure to address all these aspects in existing research hinders the development of robust solutions capable of countering emerging quantum threats [7–9]. As cryptographic frameworks face the approaching gamble of quantum assaults, the endeavors of normalizing PQC calculations that driven by associations like National Institute of Standards and Technology (NIST) assumes an urgent part in laying out an establishment for far reaching reception.

Existing surveys tend to focus either on the limited area of theoretical foundations of PQC algorithms or on evaluating their performance on general computing platforms, while neglecting the unique vulnerabilities and constraints of IoT devices. Although research on PQC is rapidly expanding, there remains a lack of comprehensive and up-to-date surveys that specifically address the vulnerabilities of IoT cryptographic protocols in the face of quantum threats and evaluate quantum-resistant countermeasures for IoT devices. These studies are either broad or narrowly focus on particular algorithms, without contrasting the vulnerabilities of classical schemes against quantum threats or defining the unique challenges and solutions applicable to IoT systems [19, 23, 41, 45, 56–59]. Key gaps include the challenge of integrating the identified schemes into existing and resource-constrained IoT environments, and insufficient analysis of real-world feasibility at scale, and a narrow scope that lacks depth investigation of cryptosystems and superficial algorithmic explanations. Furthermore, the existing research lacks adequate holistic justification against evolving quantum threats.

With large-scale quantum computers expected to be realized in near future, advancing the research and innovation conducted on PQC development is urgent, as the processes of standardization, implementation, and building confidence on deployment requires significant time frame [10–13]. In this regard, our study has wide range impacts for proactive migration toward quantum-safe cryptography which is essential for maintaining and evolving the future security of IoT infrastructures. This systematic literature review (SLR) uniquely contributes by systematically examining the specific vulnerabilities of IoT cryptographic algorithm to quantum threats and evaluating the feasibility, security, and performance of PQC countermeasures in IoT systems. Unlike previous reviews, our study focuses on synthesizing the vulnerabilities, current research trends and countermeasures, strategies, challenges, and future research needs. Critical gaps remain in achieving a secure, efficient balance and in developing robust security models against evolving quantum capabilities.

This systematic review provides a comprehensive analysis of cryptography for IoT systems in the post-quantum era. We systematically catalog vulnerabilities in current IoT cryptographic algorithms exposed by quantum threats and evaluate countermeasures to enhance protocol resilience. A key contribution is the identification and assessment of promising PQC algorithms (such as CRYSTALS-Kyber, CRYSTALS-Dilithium, and SPHINCS+) suitable for IoT devices, and supported by a novel conceptual framework and illustrated with a smart grid case study. By rigorously adhering to the PRISMA 2020

guidelines, our study makes a significant contribution through its enhanced methodological rigor. This commitment guarantees methodological transparency, enhances the reproducibility of our findings, and provides a reliable benchmark for future research. Finally, we outline critical challenges and future research directions toward offering a foundational roadmap to guide researchers and practitioners in integrating quantum-safe lightweight cryptography into IoT infrastructures.

1.2 Research question

The research questions of this systematic literature review are formulated below as follows:

RQ_1 What are the vulnerabilities of the existing IoT cryptographic algorithms that can be exploited using quantum computers? (Addressed in Sect. 4.2 (Table 5) and Discussion provided in Sect. 5.1.)

RQ_2 What are the strategies available in the literature to make the existing IoT cryptographic protocols resistant to quantum threats? (Addressed in Sect. 4.2 (Table 5) and Discussion provided in Sect. 5.2.)

RQ_3 What are the emerging post-quantum cryptographic algorithms that are suitable for IoT applications? (Addressed in Sect. 4.2 (Table 6) and Discussion provided in Sect. 5.3.)

RQ_4 What are the key challenges posed and the future prospects in quantum cryptographic protocols with respect to IoT systems? (Addressed in Sect. 4.2 (Tables 8 and 9) and Discussion provided in Sect. 5.4.)

1.3 Objective of the study

General Objective The general objective is to examine the vulnerabilities, mitigation strategies, and future directions of post-quantum cryptographic protocols in securing IoT systems.

Specific Objective.

- To explore the vulnerabilities in existing IoT cryptographic algorithms that can be exploited by quantum computers.
- To identify and examine the available strategies to enhance the resistance of the existing IoT cryptographic protocols against quantum threats.
- To identify and assess the suitability of emerging post-quantum cryptographic algorithms for IoT applications.
- To examine the challenges and future prospects in developing quantum cryptographic protocols for IoT systems.

1.4 Scope of the study

This systematic literature review aims to investigate on the existing studies that focus on IoT cryptographic protocols within the realm of quantum. The review specifically focuses on identifying the vulnerabilities in existing IoT cryptographic protocols, examining strategies and algorithms proposed to address them, and discussing the challenges and future directions highlighted in existing literature. The studies included in this

systematic review were identified using systematic searches in 9 major databases, namely are IEEE Xplore, Scopus, ACM, MDPI, Springer, ScienceDirect, Wiley, Google Scholar and Nature, using predefined keywords and inclusion/exclusion criteria. The scope of this study is defined by inclusion criteria requiring that selected works focus on IoT cryptography in the context of quantum computing, be published in English, appear in reputable journals or conference proceedings, and be peer-reviewed with clearly articulated outcomes. The temporal range was also set to include studies published between 2020 and 2025 to allow the inclusion of recent research findings. On the other hand, studies that lack a focus on IoT cryptography or/and quantum computing are excluded.

1.5 Significance of the study

The significance of this study is the intersection between the two emerging technologies, which are quantum computing and IoT cryptography. Since the development of quantum computers are continuously evolving, the existing traditional cryptographic algorithms that are safeguarding IoT devices currently will become vulnerable and the important services, industries and infrastructures enabled by these devices will also be compromised. Identifying these vulnerabilities in existing IoT cryptographic protocols and evaluating the strategies introduced in the literature to resist these quantum attacks will help to overcome the security problems. In addition, this study will provide a framework that connects IoT architecture, vulnerabilities, and PQC categories for anticipating and mitigating future security threats by identifying and evaluating the emerging post quantum cryptographic algorithms. Furthermore, this systematic review discusses the challenges and future directions in post quantum cryptography, which will offer a valuable guidance to researchers, academicians, and industries in enhancing the resilience of IoT systems against quantum-powered attacks [7].

This systematic review is organized in the following structure: Sect. 1 introduces and provides the general overview of the study, the objectives of the study, the scope and significance of the study. Section 2 presents the review of existing related literature to establish the review framework and identify the research gaps. Section 3 discusses the research methodology, which includes the explanation of the systematic approach used for the selection and analysis of relevant studies. Section 4 presents the results of the analysis and review. Section 5 dicusses the implications of the findings in detail. Finally, Sect. 6 concludes this systematic review by summarizing key findings, challenges and directions for future studies.

2 Literature review

2.1 Basics of cryptography

Cryptography can be described as the study of techniques that can be used to protect communications in the presence of security threats [8]. Its main goal is to guarantee that the encrypted data cannot be decrypted by unauthorized individuals. There are three main types of classical cryptographic algorithms, public-key cryptography (asymmetric), symmetric key cryptography, and hash functions. Symmetric based IoT Cloud environment cryptography includes popular algorithms such as Advanced Encryption Standard (AES), and Data Encryption Standard (DES), and those algorithms secure communication with the help of a shared secret key between the originator and the recipient. On the other hand, asymmetric key cryptography, such as Rivest-Shamir-Adleman (RSA),

Elliptic Curve Cryptography (ECC), and Diffie-Hellman (DH) use a public and a private key for encrypting and decrypting the data, respectively. In addition, hash functions like SHA-2 (Secure Hash Algorithm), SHA-3 are not used for encryption and decryption but rather for data integrity [9].

I. *Symmetric Key Cryptography* The symmetric key cryptographic algorithms use a single secret key shared among the communicating parties that is used for both encryption and decryption. Several studies have tried to use symmetric algorithms such as AES for IoT applications due to their good balancing between security strength and computational efficiency. In particular, AES-GCM (AES-Galois/Counter Mode) and AES-EAX have appeared as preferred choices for resource-constrained IoT environments. For example, the study in [3] explored the use of AES-128 in EAX mode with a focus on lightweight cryptography for IoT to reduce block size and key size without compromising security and simplifying rounds and key schedules. The speed and simplicity of these symmetric key cryptographic algorithms are the qualities that make them suitable for encrypting large volumes of data with limited computational resources [8]. These algorithms require less processing power and smaller memory usage compared to asymmetric cryptography, which is for IoT devices. However, secure key management and distribution is a major challenge, especially in large and dynamic IoT networks. Establishing and maintaining shared secret keys across numerous devices can be complex, and it might create vulnerabilities [10]. Key distribution and pre-shared keys can lead to huge storage overhead, particularly for large-scale distributed nodes with limited resources [11]. In addition to this, the security strength of symmetric encryption algorithms can be halved by Grover's algorithm with the utilization of quantum computers. Those problems can be mitigated by combining symmetric algorithms with quantum key distribution (QKD) protocols.

II. *Asymmetric Key Cryptography* Asymmetric key cryptographic algorithms use a pair of keys for communication, which are the public key for encryption and a private key for decryption [12]. Algorithms such as RSA and ECC are classical public-key encryption schemes. Though they are used in IoT environments, they are still considered too complex and resource-intensive to be efficiently employed in resource-constrained devices [13]. Furthermore, those classical public key encryption schemes will no longer be practically secure in the post-quantum era due to their vulnerability to quantum attacks such as Shor's algorithm. Moreover, the ECC has gained research focus for its ability to provide better security with shorter key lengths compared to RSA. To exploit this, several recent studies have focused on optimizing ECC for IoT applications, with efforts to reduce memory requirements and energy consumption. ECC-based authentication schemes and hybrid encryption frameworks that combine ECC for key exchange and AES for data encryption for IoT applications are some of the focuses of these studies [3, 8]. The authors in [3] have compared hybrid RSA/AES and ECC/AES for transport encryption, and they found that ECC/AES is highly effective for IoT-driven smart home healthcare environments by balancing security, performance, and adaptability while mitigating overheads.

III. *Key Management* The key management procedure includes the processes of key generation, distribution, storage, usage, and renewal throughout the lifecycle of cryptographic keys [14]. In IoT environments, effective key management and

distribution are crucial components for establishing secure communication and protecting data [10]. The techniques used for key generation are different depending on the algorithm and application requirements, which range from random number generation to more complex derivation processes. Key distribution methods in IoT can include configuration before deployment, secure remote provisioning, and key exchange protocols. Key renewal is also important for maintaining security throughout time by considering the potential for key compromise [14]. There are some specific challenges in ensuring scalable and secure key management in IoT environments. The vast number of devices, their diverse capabilities, with often unreliable network connectivity makes the implementation of traditional key management techniques difficult [15]. The resource-constrained nature of many IoT devices limits their ability to perform complex key management operations or securely store a large number of keys. The distributed nature of IoT systems also increases the attack surface and vulnerabilities for key compromise [10]. As a solution to those issues, modified key exchange protocols with reusable keys have been proposed [15] to address the issue of frequent key generation in resource-constrained IoT device-to-device communication. Moreover, identity-based encryption was developed by [4] and indicated as an efficient encryption tool for secure data communication in IoT because it eliminates the need for complicated certificates, which simplifies key management.

2.2 Quantum computing

While the cloud service's data storage and retrieval mechanism have exposed the issues of traditional systems, the massive data generated by multiple IoT devices is pressing for more processing power. As a result, the quantum computer can complete tasks that the classical computer is unable to. Similar to how a conventional computer manipulates its bits, quantum computers are devices that can regulate delicate quantum states. It does the calculations for hardness problems using the properties of quantum mechanics and quantum physics [16]. In comparison to traditional computers, QC can easily and swiftly complete larger operations in a matter of seconds. Because a classical computer only has two possible states, 0 or 1, it operates with bits 0 or 1, OFF/ON, or one state at a time [17]. Quantum bits, or Qubits, are what allow quantum computers to function. While the likelihood of discovering 0 or 1 is equal to 1, there is a simultaneous potential that 0 or 1, or 01 will occur [18]. Conventional public-key encryption methods hold an important role in securing communications, but are vulnerable to attacks from quantum systems. The introduction of quantum computers is presenting a major threat to the security of many commonly utilized cryptographic frameworks. Quantum computing is believed to bring both tremendous opportunities and significant threats in cryptography, a method of securing communication in the presence of adversaries. On the other hand, quantum cryptography revolutionizes secure communication by using the characteristics of quantum mechanics, offering a potentially more secure alternative to traditional cryptography, such as QKD, a key-sharing mechanism between two parties using the principle of quantum mechanics [19, 20].

2.3 Quantum cryptography

The process of manipulating and representing information in various formats, including digital, quantum, chemical, biological, structural, social, and natural, is known as information processing. Information processing and computer systems that use quantum mechanics are essential to quantum information processing. Utilizing the Heisenberg and photon polarization principles, quantum cryptography contains several protocols, including digital signatures and data storage; secure processing for multiple users, quantum-powered key distribution and entanglement and quantum-enabled random key generator [18].

2.4 Quantum threats on cryptography

In the 1990s, the development of Shor's and Grover's algorithms established the foundation for understanding how quantum computing could challenge traditional cryptographic systems. The threat by quantum algorithms arises from the advanced computational power of quantum computers to solve specific mathematical problems exponentially faster than classical systems. These threats primarily undermine the core hardness assumptions of public-key cryptography, including integer factorization, discrete logarithms, and elliptic-curve discrete logarithms [21]. For example, Shor's algorithm, which efficiently solves the cryptographic schemes relies on integer factorization and discrete logarithm problems. Similarly, Grover's algorithm effectively reduces the security of symmetric ciphers like AES quadratically. In contrast to other vulnerabilities, quantum threats do not only exploit algorithmic weaknesses or implementation flaws but fundamentally challenge the algorithms themselves when scalable quantum computers become available, enabling strategies like "harvest now, decrypt later," (HNDL) where encrypted data that are collected at one point in time can be decrypted in the future.

Classical vulnerabilities involve cryptanalytic attacks carried out on traditional computers, targeting mathematical flaws, inadequate key sizes in cryptographic algorithms. Examples include brute-force attacks, which exhaustively test all possible keys space, requiring infeasible time on classical hardware as well as differential or linear cryptanalysis on block ciphers and exploitation of structural weaknesses in systems like certain multivariate polynomial schemes. These attacks are constrained by the polynomial or exponential time complexity of classical computing, rendering them ineffective against robust, well-parameterized algorithms.

Implementation-based vulnerabilities rise from faults in the practical deployment of cryptographic systems, including software bugs, hardware weaknesses, and configuration errors, rather than from inherent problems in the mathematical algorithms. Examples include side-channel attacks such as timing or power analysis that reveals key information during computation and faulty random number generation that result in predictable keys and insecure key management in IoT devices. These vulnerabilities can compromise even theoretically secure algorithms by exploiting how the algorithm is executed in real-world environments. This kind of flaws can be mitigated using secure implementation techniques, constant-time implementations or hardware security modules.

I. *Impact of Shor's Algorithm* Shor's algorithm, which introduced by Peter Shor in 1994, utilizes quantum computers to solve the discrete logarithm (DL) and integer

factorization mathematical difficulties in polynomial time exponentially faster than classical methods. By employing quantum Fourier transforms, it efficiently finds the period of a function, enabling the rapid factoring of large numbers [19]. The algorithm presents a considerable security risk to the traditional cryptographic methods. This ability threatens the integrity of standard searchable encryption methods and cryptographic systems that depend on DL and integer factorization. It poses a significant security risk for the highly used public-key cryptosystems, including RSA, ECC, DH and ECDH key exchange, potentially making them ineffective [10, 19]. In particular, cross-domain Direct Anonymous Attestation (DAA) protocols utilizing ECC or RSA encryption, employed for authentication across various trust domains in IoT, are at risk from quantum computing attacks.

Shor's Algorithm can break RSA by factoring the large composite modulus such as, 2048-bit numbers into its prime factors, allowing the recovery of the private key in polynomial time. This makes RSA insecure in the presence of scalable quantum computers. It compromises elliptic-curve cryptography by solving the discrete logarithm problem on elliptic curves, enabling attackers to derive private keys from public keys, thus making ECC vulnerable. However, it has no significant effect on symmetric cryptography like AES and hash functions like SHA-families as they do not rely on the problems solved by Shor's Algorithm.

II. *Impact of Grover's Algorithm* The Grover's algorithm was initially developed by Lov Grover in 1996, poses a significant challenge to symmetric key cryptographic systems by offering a quadratic enhancement for brute-force search problems on quantum computers. Specifically, it enables the identification of a specific item in an unsorted database of N items in approximately $\sqrt{N}$ steps, dramatically reducing the effort required for exhaustive search. It also can be used to brute-force key searches and finding hash pre-images [21, 23].

In opposite to Shor's algorithm, which directly undermines public-key systems, Grover's algorithm effectively halves the security strength of symmetric encryption. A 256-bit key in a symmetric encryption algorithm like AES can provide only 128 bits of security against an attacker using a quantum computer equipped with Grover's algorithm [21]. This acceleration also undermines cryptographic hash functions such as the SHA-2 and SHA-3, which play critical roles in digital signatures, authentication, and data integrity [9, 22]. In the realm of IoT, where symmetric encryption and hash-based techniques are commonly utilized for their efficiency and minimal computational demands, the ramifications of Grover's algorithm are especially significant. Systems that depend solely on 128-bit or shorter symmetric keys could become susceptible to practical quantum attacks, prompting a necessary transition to longer key lengths and quantum-resistant options. Additionally, lightweight cryptographic schemes tailored for resource-limited IoT devices must now consider quantum adversaries, adding complexity to the development of secure and efficient protocols. Therefore, Grover's algorithm underscores the importance of strong symmetric key strategies and indicates the

pressing necessity of moving to quantum-resistant standards, even for systems that do not utilize public-key cryptography [23].

Grover's Algorithm allows a quadratic speedup for search problems on quantum computers by reducing the time to search an unsorted database of N items dramatically. In symmetric cryptography, such as AES [172], this algorithm influences exhaustive key searches by effectively halving the security level of keys. For example, AES-256, which in the classical case requires 2^{256} operations, reduced to about 2^{128} quantum operations, while AES-128 minimized to approximately 2^{64} [21]. Therefore, it is critical to note that Grover's algorithm effectively halves the security level of symmetric keys (e.g., AES-256 provides ~ 128 bits of post-quantum security), requiring larger key sizes but not rendering them completely obsolete.

III. *Current Post-Quantum Cryptographic Solutions* Due to the vulnerability presented by Grover and Shor's algorithm, QKD and PQC were designed as cryptography approaches aimed at securing digital communications against the growing threat of quantum computers. Although they are introduced by the vulnerability introduced by quantum algorithms, they differ significantly in their implementation feasibility and adaptability. QKD is a hardware-based quantum cryptography technique that uses the principles of quantum mechanics to allow two parties to generate and share a secret key [24]. The most well-known protocol in this space is BB84, which employs quantum bits (qubits) sent through quantum channels to identify any attempts at eavesdropping [14]. QKD works based on the principles of physics, specifically the no-cloning theorem, which states that perfectly duplicating an unknown quantum state, is impossible [25]. As a result, any attempt to intercept the key exchange would create noticeable anomalies, enabling parties to reject compromised keys. QKD is a quantum cryptography utilized to produce a common key, which is subsequently used to protect the symmetric keys through One-Time Pad (OTP) encryption [26]. There are two essential elements of quantum mechanics that the success of quantum cryptography relies on: The Photon Polarization Principle and the Heisenberg Uncertainty Principle [27]. A practical QKD technique was simulated with up to 7500 photons, examining the relationship between transmitted photons and key length. In contrast to binary simulations, it accurately represents real photon polarizations, improving both randomness and adaptability [28]. QKD focuses mainly on the challenge of securely exchanging keys, whereas PQC provides a wide range of cryptographic capabilities, such as digital signatures, public-key encryption, and key encapsulation. This wider range of functions gives PQC an advantage in safeguarding not just communication channels but also data storage, software updates, and methods of authentication.

PQC is a software-focused approach that involves creating new mathematical algorithms that can resist both classical and quantum threats. Unlike QKD, PQC operates without the need for quantum hardware or channels, making it suitable for implementation on current classical infrastructures. This method aims to substitute public-key algorithms at risk such as RSA and ECC with alternatives grounded in lattice issues, coding theory, multivariate expressions or hash functions [29]. These algorithms are crafted

to be secure against established quantum attack methods while maintaining compatibility with existing systems and protocols. As IoT influences the Internet's future, cryptographic techniques like public-key encryption (PKE) must be used to safeguard communications between nodes. Nevertheless, traditional PKE techniques like ECC and RSA have great complexity and are susceptible to quantum attacks. Because post-quantum schemes based on the learning with errors (LWEs) problem are less complex than PKE schemes, they have attracted a lot of attention in the last 10 years [12].

2.5 Categories of post-quantum cryptography

I. *Lattice-based cryptography* Lattice-based cryptography is one of the most promising and extensively studied branches within PQC, due to its combination of strong security assurances and efficient computational performance. It achieves security and adaptable access control mechanisms for multi-user applications by utilizing the learning with errors (LWE) assumption with the combined public key (CPK) technique [30] and subset predicate encryption [8]. The most widely used encryption in the NIST PQC standardization program is based on the Ring Learning with Errors (RLWE) lattice. RLWE has been optimized to create a lightweight Circular Learning Error Algorithm (CELA) that is suitable for IoT- cloud environments [31]. CRYSTALS-Dilithium and CRYSTALS-Kyber are other lattice-based methods that are notable digital signature schemes known for their simplicity and efficiency [32]. Falcon offers compact signatures, making it ideal for systems constrained by bandwidth [32]. Kyber has garnered attention for its robust post-quantum security and minimal key sizes, serving as a key encapsulation mechanism (KEM) [33, 34]. The Ring-BinLWE is a type of lattice-based cryptography suitable for resource-limited IoT devices [35]. Furthermore, Number Theory Research Unit (NTRU) a public key encryption and simple implementation lattice-based encryption method [36], remains relevant due to its lightweight characteristics, making it particularly fitting for resource-constrained IoT devices and Vehicular Ad hoc Network (VANET) [37, 38]. The modifications and adaptations of established cryptographic standards, specifically RSA-2048, RSA-7680, NTRU-401, NTRU-593, SABER, and LightSABER explored in [1]. NTRU and SABER are lattice-based PQCs that offer efficient encryption with low computational overhead. However, these methods impose limitations on the size of the data that can be directly encrypted. For 128- bit security, both NTRU and SABER are restricted to encrypting only 16 bytes of data, and for 196-bit security, this limit extends to 32 bytes. Implementing these lattice-based PQCs in a constrained IoT device is challenging. To tackle these issues, this paper introduces an effective processor for lattice-based cryptography that features specialized Single- Instruction Multiple-Data (SIMD) instructions [39].

II. *Code-based cryptography* Code-based cryptography is another PQC that is a potential successor to current public key systems, relying on the difficult problem of decoding unknown error-correcting codes and offering strong security. Algorithms like McEliece and Niederreiter/Xing utilize large keys, posing implementation challenges for resource- constrained devices [40]. Encryption involves encoding messages with added errors, while decryption recovers the original message by removing these

errors. Securing the underlying code structure, ideally by disguising it as a generic code, is crucial for preventing attacks [41].

III. *Multivariate polynomial cryptography* Multivariate polynomial cryptography is another PQC focusing on the difficulty of solving systems of multivariate quadratic equations over finite fields. Rainbow was once a strong candidate in the NIST competition recognized for its efficient signing process and relatively short signatures [42].

IV. *Hash-based cryptography* Hash-based cryptography secures through the cryptographic hash functions and is mainly utilized for digital signatures. This category is considered to be highly secure, given its solid foundations that are minimally impacted by quantum algorithms. SPHINCS + is a stateless hash-based signature scheme that ensures strong post-quantum security by integrating multiple hash-centric approaches [43, 44]. In contrast, XMSS (eXtended Merkle Signature Scheme) operates as a stateful scheme that has already been standardized and is well-suited for situations that require long-term signature capabilities with verifiable security assurances. Hash-based signatures, post-quantum technique, built on one-time signatures (OTS). OTS's vulnerability to key reuse led to Merkle trees, which use hash trees to authenticate OTS public keys. The Merkle signature scheme's public key is the tree's root, and the private key is a set of OTS secret keys, potentially generated from a seed to save storage. Key reuse is prevented by ordered leaf node usage [41].

V. *Isogeny-based cryptography* Isogeny-based cryptography is a more recent advancement in PQC, grounded in the difficulty of identifying isogenies such as mathematical mappings between elliptic curves [35]. This category has gained attention for its exceedingly small key sizes, which are attractive for IoT devices. Super-singular Isogeny Diffie-Hellman (SIDH) and its variant, Super-singular Isogeny Key Encapsulation (SIKE), were promising schemes in this area. However, recent advancements in cryptanalysis have significantly diminished their perceived security, resulting in a decrease in confidence in their utility for secure communications [45].

2.6 Internet of things

IoT contains a vast network of interconnected machines with embedded software and hardware designed for large-scale cooperation and communications. Low-power IoT devices (also sensors) are typically based on low-cost microcontrollers interconnected via low-power radio or wireless communications [170, 171]. They offer a different trade-off such as smaller capacity in computing, networking, and memory, in order to achieve radically lower energy consumption. IoT connectivity stacks combine LoWPANs (low-power wireless personal area networks) for physical links, the RPL (Routing Protocol for Low-power and lossy networks) routing protocol for data exchange, and lightweight application protocols namely MQTT (Message Queue Telemetry Transport) and CoAP (Constrained Application Protocol). This interconnectedness has presented unique security challenges due to the resource constraints of the devices and their physical availability in distributed and hostile environments [14, 173]. The integration of IoT devices into various applications has brought about significant advancements, but it also introduces new security challenges [174–177]. Cryptographic protocols are essential to mitigate these security challenges and ensure data confidentiality, integrity, authentication, and

privacy in the IoT system [12]. These protocols use various techniques to protect the data during collection, transmission, and also processing in the IoT nodes. In addition to the limitations of IoT devices, the forthcoming security threats from quantum computing are dictating careful consideration in the selection and design of these cryptographic algorithms [46].

2.7 Emerging IoT cryptographic approaches

I. *Lightweight PCQ* The resource limitations of many IoT devices have led to the development of lightweight cryptographic protocols specifically designed for low-power, low-memory, and low-computational power devices. These algorithms' target is to provide adequate security with minimized resource consumption [46]. Example, these include lightweight block ciphers [3], lightweight hash functions [47], and optimized versions of existing algorithms [48]. The ring-binary learning with errors (Ring-BinLWE) is a lattice-based cryptographic scheme that has gained scholars' attention for its simplicity, which makes it suitable for IoT applications. Implementations of resource-constrained microcontrollers such as the AVR ATxmega128A1 and the ARM Cortex-M0 have also been explored in some studies [12]. In addition, a new multiplication method called in-place modular reduction and anti-circular rotation column-based multiplication has been proposed to optimize the efficiency of Ring-BinLWE for resource-constrained devices [13]. Post- Quantum Cryptography The rapid progress in developing quantum computers is becoming a significant security problem for traditional cryptographic systems, such as RSA and ECC, which are used in securing the majority of communication channels nowadays [49]. Quantum computers are believed to potentially break those widely used traditional encryption algorithms [50]. Therefore, the timely adoption of post-quantum cryptographic algorithms that are robust against quantum attacks is highly important for IoT security. Lattice-based cryptography, especially Learning with Errors (LWE) and its variants, requires the lowest computational complexity among current post-quantum PKE schemes, which makes them more suitable for resource-constrained IoT devices [48].

Recently several researchers have explored PQC schemes, including lattice-based mechanisms such as Ring-BinLWE [48], NewHope, FrodoKEM, and CRYSTALS-Kyber [38], isogeny-based cryptography like Supersingular Isogeny Diffie-Hellman (SIDH) [51]; code-based cryptography like McEliece [3], and multivariate quadratic cryptography. Implementations of Ring-BinLWE on RISC-V platforms show speed improvements. The NIST has endorsed CRYSTALS-Dilithium as a leading lattice-based digital signature algorithm [47]. Other research also focuses on lightweight hardware accelerators for post-quantum digital signatures like CRYSTALS-Dilithium [47]. In other findings [30], lattice-based IBE schemes are mentioned as extremely suitable for fog computing due to their resistance to quantum attacks and simplified certificate management. Hybrid cryptography combines the strengths of different cryptographic techniques, mostly integrating classical algorithms (like symmetric encryption) with emerging or more robust methods (like asymmetric or post-quantum encryption) [26]. For example,

a hybrid encryption framework might use ECC for key exchange and AES for efficient data encryption, addressing performance and near-term security concerns. Another essential and timely issue is combining classical and quantum-resistant cryptographic algorithms to provide immediate security solutions and a transition path to a quantum-safe future [46]. In addition, within the context of searchable encryption for IoT, homomorphic encryption can be combined with other techniques to enhance retrieval capabilities and security [52].

II. *Hybrid post-quantum cryptography* Hybrid post-quantum cryptography has surfaced as a transitional method for securing communication in the upcoming post-quantum era. In these hybrid schemes, conventional cryptographic techniques like RSA or ECC are merged with KEM to protect against both classical and quantum threats [53]. This dual-layered security is especially beneficial during the transition phase when quantum risks are expected but not yet widespread. Hybrid methods enable systems to remain compatible with existing frameworks while incorporating quantum-resistant elements like NTRU and RSA [54]. They are particularly appealing for IoT systems that necessitate both backward compatibility and security that can withstand future quantum threats, supporting a gradual transition without compromising current performance or interoperability [55]. Emphasizes the hybrid combination of quantum teleportation protocol and remote state preparation protocol into one interconnected communication framework in an IoT context that is used together over a shared quantum channel, specifically a six-qubit entangled state. These two-protocol combinations are not PQC but emphasize the use of more than one protocol to make them more secure cryptographic protocols. In studies such as in [30] findings, lattice-based Identity based encryption (IBE) schemes are mentioned as extremely suitable for fog computing due to their resistance to quantum attacks and simplified certificate management. In addition, within the context of searchable encryption for IoT, homomorphic encryption integrated with other techniques used to enhance retrieval capabilities and security [52].

III. *Standardization effort* The NIST PQC competition for standardizing quantum-resistant cryptography has a shortcoming (a mixed and largely problematic record) when its suitability assessed for certain IoTs [62]. On the other hand, the selected key encapsulation mechanism, CRYSTALS-Kyber (ML-KEM), shows promise with its relatively small keys and fast operation, the signature schemes present significant hurdles. Standards like CRYSTALS-Dilithium (ML-DSA) and SPHINCS + generate large signatures and have high computational costs that are often prohibitive for resource-constrained IoT devices [31]. Furthermore, the competition rightly filtered out candidates with extreme overheads, such as Classic McEliece with its megabyte-sized keys. In essence, NIST prioritized general-purpose security and efficiency, leaving a critical gap for the most constrained IoT environments, where the standardized algorithms' memory, power, and bandwidth demands remain a major challenge [21].

2.8 Related work

The quantum threat landscape is a clear and present danger to long-term data security where a proactive necessitates and system-wide migration to PQC to build resilient systems capable of withstanding this future computational shift. In this work, we identifies critical challenges and future research trajectories in the development of quantum-resistant cryptography for IoT, and thereby guiding researchers and practitioners in fortifying IoT infrastructures against the emerging threat of quantum computing. Table 1 summarizes similar studies on post-quantum cryptography for IoT, comparing their objectives, methods, findings, strengths, and limitations.

3 Research methodology

This SLR study was conducted using the Preferred Reporting Items for Systematic Reviews and Meta-Analysis (PRISMA 2020) as a guideline [60]. Adhering to this guideline ensured methodological consistency, transparency, and reproducibility throughout the research process and among the researchers. As a result, a cohesive and transparent workflow was designed to produce reliable and evidence-based insights while adhering to scientifically recognized standards for systematic reviews.

3.1 Systematic literature review process

As it is depicted below in Fig. 1, the research processes of this study are organized into the following three phases:

I. *Phase I - Research Directives* The foundations for the systematic review are established at this phase by defining the objectives, scope, and methodology.

- *Protocol Definition* At the beginning of this study, a detailed work plan and guidelines that outlined the processes to conduct the systematic review were prepared. It includes specifying objectives, scope, methods, and criteria for the selection of the studies. These established guidelines or protocols helped as a roadmap for the research team and ensured transparency, consistency, and reproducibility throughout all the processes of the systematic review.
- *Decision Literature Survey* This step focuses on conducting a more general survey of the existing literature that is related to this research topic. It mainly emphasized on identifying key studies, theories, frameworks, and gaps in the domain knowledge. This survey helps to determine the research direction by providing ideas into what has been studied and where further exploration is needed. At this stage, additional clarity was gained from analyzing the surveyed literature, which enabled the researchers to sharpen their focus and define precise objectives.
- *Research Questions Definition* Based on the directions obtained from the literature survey and protocols, specific research questions are formulated. These questions guided the systematic review by helping to address the targeted problems and set the boundaries of the systematic review by indicating what will be covered and excluded.

II. *Phase II - Data Collection* This phase includes searching and filtering relevant studies to gather data for analysis.

Table 1 Summary of related work on Post-Quantum cryptography for IoT

Ref.	Objective	Methods	Findings/contributions	Strengths	Limitations / gaps
[23]	To assess the efficiency and feasibility of implementing post-quantum cryptographic algorithms on constrained IoT devices, like Raspberry Pi 4	Performance evaluations using Raspberry Pi 4 devices, Analysis of key generation, encapsulation, and verification signature	CRYSTALS-Kyber and CRYSTALS-Dilithium are identified as the most efficient schemes for key encapsulation and digital signatures, respectively	The article provides a comprehensive performance evaluation designed to simulate IoT devices.	Integration of the studied schemes into existing systems poses a hurdle due to compatibility issues
[56]	To analyze post-quantum cryptographic schemes for securing authentication and user privacy in IoT devices.	Literature review, Analysis, and evaluation of post- quantum cryptographic schemes	Emphasizes the essential role of PQC in safeguarding IoT networks from quantum attacks, with a focus on suitable lightweight cryptography	Provides valuable insight on analysis and evaluation of post-quantum cryptographic schemes in IoT devices	The challenge of implementing post-quantum cryptography in resource-constrained IoT environments
[19]	To explore the intersection between quantum computing and the security challenges created by IoT applications analyze the security concerns of IoT smart applications and survey quantum-based solutions	Literature review, Security analysis related to IoT applications	Quantum computing principles like superposition and entanglement can en- able secure communication; QKD is identified as a promising solution for securing IoT communication.	Provides a comprehensive survey covering both the theoretical foundations of quantum computing and its practical impacts on IoT security.	Limited focus on implementation details, practical feasibility of deploying QKD at scale is not thoroughly analyzed
[57]	To survey post-quantum public-key signature schemes for securing vehicular communications, To provide guidelines for adapting suitablepost-quantum candidates	Literature review of post- quantum signature schemes, Comparative analysis of variouspost-quantum cryptographic approaches	Identification of five main directions of PQC approaches: multivariate-based, hash-based, lattice-based, code-based, and super-singular isogeny-based. A hybrid scheme for transition is suggested	Provides a clear transition strategy guideline	Lack of experimental validation of the schemes, Finding optimal balance between security, performance and resource requirements remained an open challenge
[58]	Evaluating and recommending post-quantum cryptographic algorithms to secure IoT devices against quantum computing threats	Analysis and comparison of different post-quantum cryptographic algorithms, Experimentation	Kyber is a promising post-quantum cryptographic algorithm for IoT security, Performance comparisons of Kyber and BIKE on different platforms	Provides a comparative analysis of relevant post-quantum algorithms, Provision of empirical performance data	The compromise of the SIKE algorithm highlights the necessity of additional research.

Table 1 (continued)

Ref.	Objective	Methods	Findings/contributions	Strengths	Limitations / gaps
[45]	To analyze the performance of selected PQC algorithms specifically for resource-constrained IoT devices	Empirical analysis performance evaluation of KEM, Comprehensive analysis	Provides a guideline - for selecting appropriate cryptosystems, LightSaber exhibits efficient memory usage while Kyber512 demonstrates the best CPU performance	Provides practical guidelines enabling practitioners to select suitable post-quantum crypto- graphic mechanisms based on device capabilities	Lack of depth in its explanation of the algorithms discussed and the scope of cryptosystems examined is quite limited
[59]	To explore PQC techniques as a solution for securing resource-constrained IoT devices against quantum threats	Literature review, identification of IoT security issues, analysis of PQC standardization efforts	Lattice-based cryptography being the most promising PQC solution, PQC implementations face practical challenges in IoT environments	Provides a comprehensive review of IoT security issues	Lack of practical implementation and performance evaluation
[41]	To investigate the vulnerabilities level of current IoT protocols	Literature review, case studies, comparative evaluation of existing 3GPP IoT solutions and frameworks	Identification of potential threats for the majority of IoT implementations – including DoS, DDoS, brute force attacks, direct channel attacks and injections	Provides a thorough evaluation of existing vulnerabilities in current IoT security frame- works	Current security models are potentially insufficient to address the evolving capabilities of quantum computing and the specific limitations of IoT devices

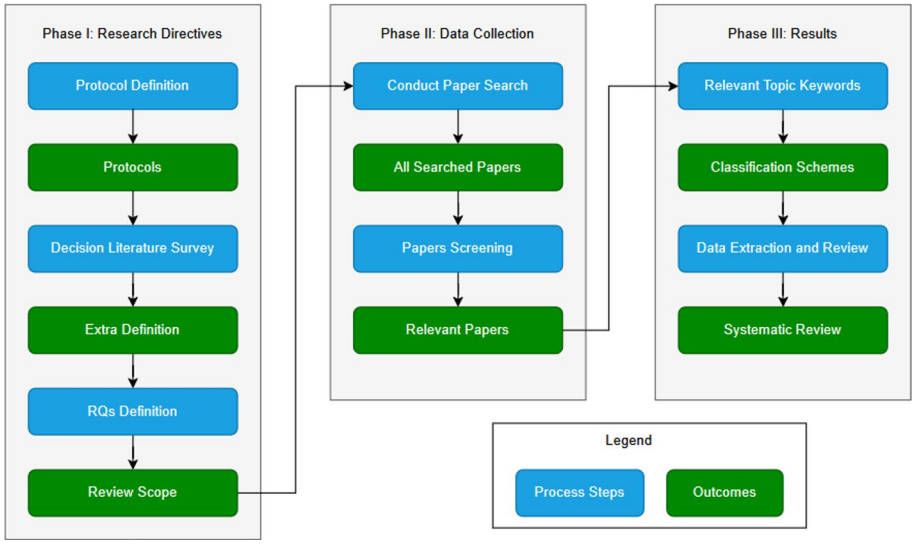


Fig. 1 Systematic literature review process used in this study

- *Conduct Paper Search* At this stage, the researchers performed a search across all over the databases using predefined keywords, inclusion/exclusion criteria, and search strategies that are defined in the protocol. This step was aimed at collecting all potentially relevant studies for initial consideration and further screening. Efficient

queries were prepared and executed in nine relevant databases, namely IEEE Xplore, Scopus, ACM, MDPI, Springer, ScienceDirect, Wiley, Google Scholar, and Nature. The queries were carefully formulated to access a broad spectrum of relevant literature for the study.

- *Papers Screening* The collected papers went through several steps of screening to determine their relevance based on criteria such as their topic's alignment with the scope of the study, methodological quality, and relevance to the research questions. The screening process was conducted in multiple stages, resulting in exclusion of irrelevant and low-quality papers from further analysis and review. It began during the initial the search for the relevant articles by adding different filters to the query to narrow down the search results and retrieve the relevant papers. A title and abstract-based screening was conducted to further minimize the number of papers and identify the most relevant studies. In addition to these, a full-text evaluation was conducted to confirm the eligibility of the papers for the scope of the study.

III. *Phase III - Results* This phase mainly emphasized on analyzing the selected papers, synthesizing the findings, and identifying challenges and future directions.

- Relevant Topic Keywords: Keywords from the relevant and selected papers were extracted to classify the ideas into different topics and subtopics within the scope of the study. This step organized the information into categories, which made the synthesis and reviewing easier.
- Data Extraction and Review: Finally, the researchers extracted detailed data from the selected papers using an extraction form. This template was used to systematically extract relevant information from the papers, including the study's characteristics, methodologies, and outcomes. The extracted data is then reviewed comprehensively to synthesize findings. This process ensures that all relevant evidence is considered systematically and helps facilitate a uniform and consistent way of comparing the results and analyzing the findings of the selected papers.

3.2 Data collection

I. *Data Source* The search considered nine databases, which were selected carefully due to their massive collection of articles that focus on computer science, cryptography, and quantum technologies and their indexing of high-impact journals and conference proceedings. Figure 2 shows the databases considered in this study and the number of articles obtained with the initial search query.

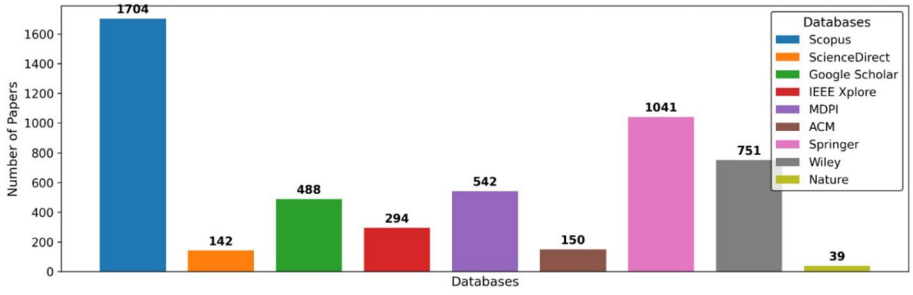


Fig. 2 Databases with their initial search results

II. *Search String and Strategy* The search strategy was conducted systematically and numerous relevant papers were obtained from nine (9) reputable databases. These databases are IEEE Xplore, Scopus, ACM, MDPI, Springer, Sci- enceDirect, Wiley, Google Scholar and Nature. The advanced search option was used in each database to execute a more appropriate search query and obtain a more comprehensive result. To retrieve relevant literature from these databases, the following key search terms were used in various combinations: "IoT", "Internet of Things", "Cryptography", "Encryption Scheme" and "Quantum". Using these keywords, the search query (string) was formulated as: ((("IoT" OR "Internet of Things") and ("Cryptography" OR "Encryption Scheme")) AND Quantum). However, since each database has slightly different rules for a search query, this generic query was customized for each database and used accordingly as stated in Table 2.

From the first search, a total of 5151 papers were obtained from these nine (9) databases. Furthermore, query filtering was applied, including publication date (2020–2025), language (English) and article type (journal and conference). The query optimized using this query filtering approach yielded a total number of 1157 more relevant papers. The metadata of these papers were downloaded and imported into Zotero referencing tool for identifying duplicates. The previous studies have been reviewed comprehensively in earlier surveys; therefore, our review targets to avoid redundancy and instead focused on the intersection of IoT, Cryptography, and Quantum Computing. Duplicates literature sources were identified and removed using Zotero, a reference management tool, to ensure the accuracy and integrity of the research database. As a result, a total of 78 duplicate papers were removed and the remaining 1079 papers were considered for further screening Fig. 3.

III. *Inclusion and Exclusion Criteria* After inserting the search query in each database, the following inclusion and exclusion criteria were applied as stated in Table 3.

IV. *Study Screening and Selection* The total number of papers (1,079) obtained after deduplications were further screened by examining their title and abstracts. The papers were distributed to two research team members, and the screening was conducted independently. This helped to critically and swiftly eliminate irrelevant or low-quality articles. Through this process, a total of 164 papers were deemed eligible for full-text retrieval and review.

Table 2 Adopted search string for each database

Database	Search string used
Scopus	TITLE-ABS-KEY ('IoT' OR 'Internet of Things' AND 'Cryptography' OR 'Encryption' AND 'Quantum' OR 'Post-quantum' OR 'Quantum AND Computing')
ScienceDirect	("IoT Cryptography") OR ("Internet of Things cryptography") AND quantum
Google Scholar	Title("IoT cryptography" OR "Internet of Things cryptography" AND quantum)
IEEEXplore	("All Metadata": IoT) AND ("All Metadata": Cryptography) AND ("All Metadata": Quantum)
MDPI	Abstract((IoT OR "Internet of Things" OR "Resource-constrained devices") AND (Cryptography OR "Encryption scheme") AND(Quantum OR Post-Quantum))
ACM	IoT AND Cryptography AND Quantum
Springer	("Internet of Things" Cryptography AND Quantum) AND (content-type:"Article" OR content-type:"ConferencePaper")
Wiley	("IoT" OR "Internet of Things") AND "cryptography" AND "quantum computing"
Nature	IoT AND Cryptography AND Quantum

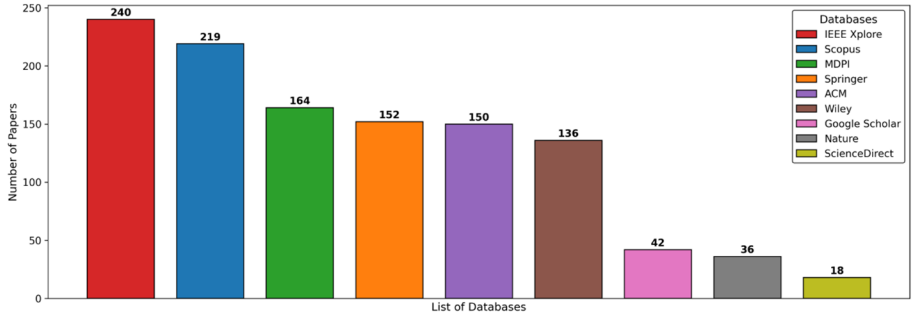


Fig. 3 Number of papers obtained using the optimized query

Table 3 Inclusion and exclusion criteria

Criteria type	Details information with examples
Inclusion	Papers relevant to the defined research objectives
	Literature published between 2020 and February 12, 2025
	Peer-reviewed articles only
	Articles published in only English
	Journal and conference papers included
Exclusion	Articles outside the defined research scope
	Papers published before 2020 or after February 12, 2025
	Non-peer-reviewed papers
	Papers published in non-English languages

- *Full-text Screening* Finally, the 164 papers accepted through title and abstract-focused screening was further considered for full-text screening to ensure their relevance to the research questions and objectives of the study. Among these papers, the full text of one (1) paper was not retrieved, and two (2) papers were excluded due to the irrelevance of their content in answering the research questions of this study. Eventually, a total of 161 papers were accepted and reviewed in this study using our data extraction form. A clearly labeled full PRISMA 2020 flow diagram with all required stages (identification, screening, eligibility, and inclusion) and the corresponding numbers from our study was shown in Fig. 4.

3.3 Data extraction process

A well-structured data extraction form was prepared and used as a framework for systematically collecting key information from the selected studies that are included in the systematic literature review (SLR). As shown in the supplementary section, Appendix A (Table A.1), the form helps to capture bibliographic details such as the article title, author(s), publication year, journal/conference, and DOI for reference management. Additionally, it documents research-specific elements, including objectives, research methodology used, key findings, and discussions on cryptographic protocols and their vulnerabilities to quantum threats. Furthermore, the form includes elements that focus on identifying which research question is answered by the reviewed paper. The research questions focus on identifying the vulnerabilities in existing IoT cryptographic protocols, mitigation strategies and their effectiveness, the cryptographic schemes and their suitability for IoT environment, the challenges, future research directions, and also emerging trends in quantum cryptography for IoT devices. This structured approach was used to retrieve more complete and logically consistent information and analysis of

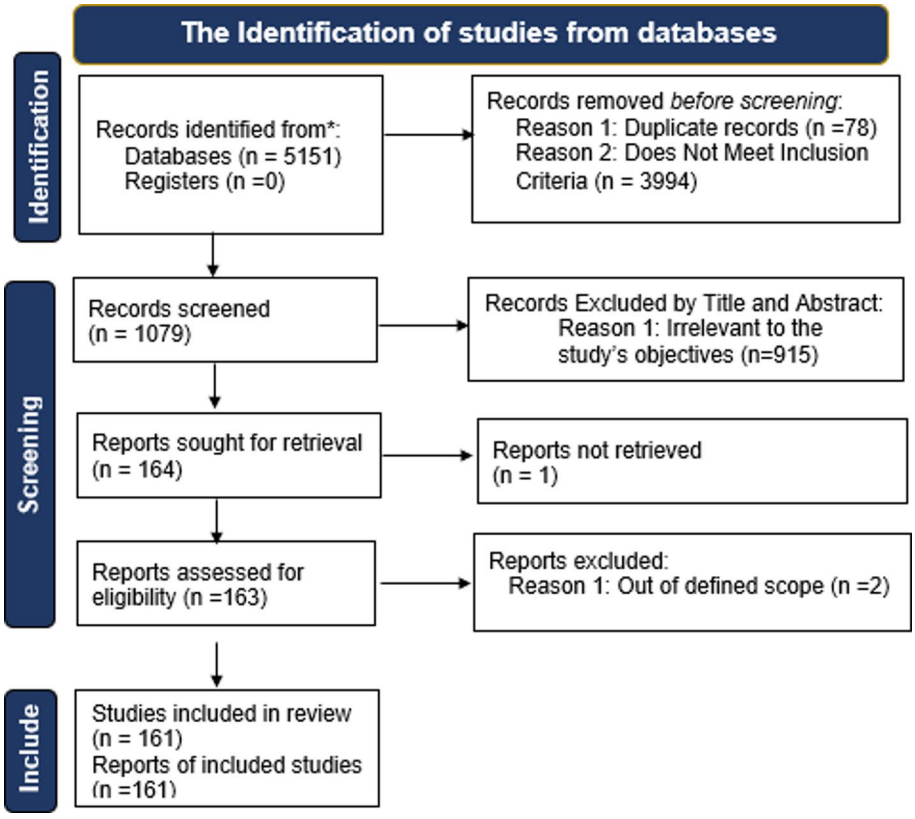


Fig. 4 PRISMA 2020 flow diagram illustrating the relevant papers identification, screening, eligibility, and inclusion of the study

security concerns in IoT cryptography. Furthermore, the data extraction form contains elements that assess the strengths of each study, limitations or weaknesses, and potential research advancements. This systematic and structured framework was used to ensure that the literature review remains thorough, well-organized, consistent, and aligned with the research objectives, thereby supporting effective knowledge synthesis and guiding future research directions in IoT security.

3.4 Quality assessment criteria

To ensure the reliability and validity of the selected studies, a quality assessment was conducted using the PRISMA guidelines and predefined criteria. These criteria were adapted from well-known guidelines in systematic reviews and focused on relevance to research problems and objectives of post-quantum cryptographic protocols in IoT, methodological soundness, clarity of research objectives, and significance of the findings and transparency.

Then, we have outlined the six definite criteria used to evaluate the quality of the included studies where each criterion was rated on a 1–5 scale shown in Table 4. Each study was evaluated based on these criteria to determine its inclusion in the systematic review. The steps followed to obtain articles with higher quality includes, selecting the most reputable databases, defining inclusion and exclusion criteria, assessing the quality of each paper, documenting the quality assessment process and incorporate the findings in the analysis process. Finally, only studies scored an average of 3.5 or higher across the criteria were included in the full text review. The screening was conducted by two

Table 4 Quality assessment criteria

Criterion	Description	Scoring scale (1–5)
Clarity of research objectives	Are the research goals clearly stated and aligned with the study's scope?	1 = unclear, 5 = very clear
Methodology Validity	Are the methods appropriate, rigorous, and reproducible?	1 = weak, 5 = robust
Relevance to PQC in IoT	How directly does the study address post-quantum cryptography in IoT contexts?	1 = low relevance, 5 = highly relevant
Transparency of Results	Are the findings clearly presented and supported by data or analysis?	1 = vague, 5 = detailed and supported
Relevance to review question	Degree to which findings inform the review objective	1 = Low relevance; 5 = High relevance
Publication Quality	Peer-reviewed publication in a reputable venue	1 = unknown/low-quality source, 5 = high-impact peer-reviewed source

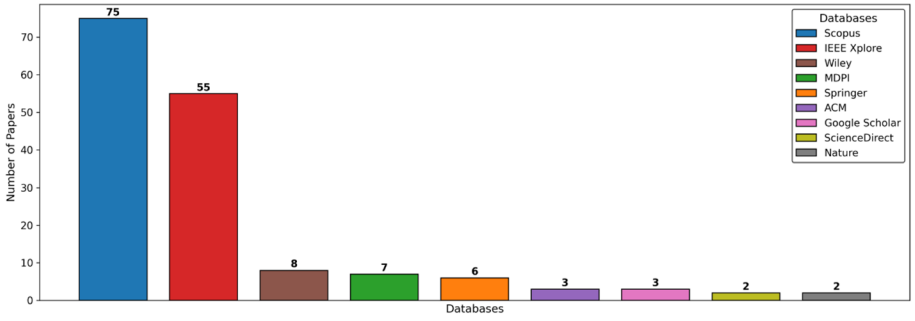


Fig. 5 Number of accepted papers

independent reviewers based on the literatures titles and abstracts. However, prior to the full screening, a pilot screening of 30 studies titles and abstracts was conducted to calibrate the process. The Cohen's κ was calculated to assess inter-rater reliability and it produce $\kappa = 0.84$ that indicates substantial agreement Fig. 5.

4 Results

This section presents the findings of study in two parts: descriptive statistics that quantify the core data, and a thematic analysis that identifies and explores the key themes, challenges, and trends from the literature.

4.1 Descriptive statistics

The quantitative summary of the dataset is presented.

I. *Article Categorization by Year* Fig. 6 shows the number of papers published each year from 2020 to 2025. Thus, most papers were published in 2022 and 2024. Specifically, the year 2024 has the highest percentage of papers, indicating an increase in research activity within this time frame. A few papers were included in those selected databases in 2025 (until February 12, 2025) indicate that the data for this year is still in the process of being assembled.

II. *Publication Types* Fig. 7 presents the number of articles reviewed categorized by publication type. Out of all the reviewed papers, 61.49% were published in journals, while 38.51% were from conferences. This means most of the reviewed studies were published in journals.

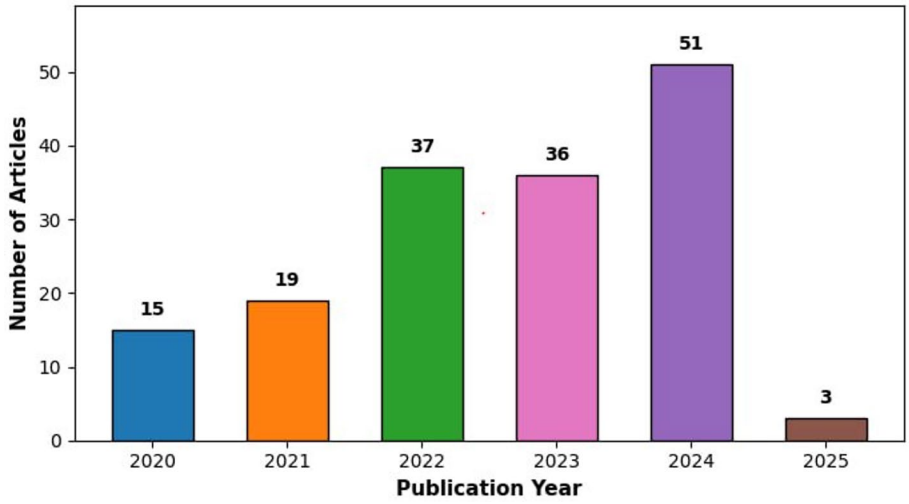


Fig. 6 Categories of articles by years

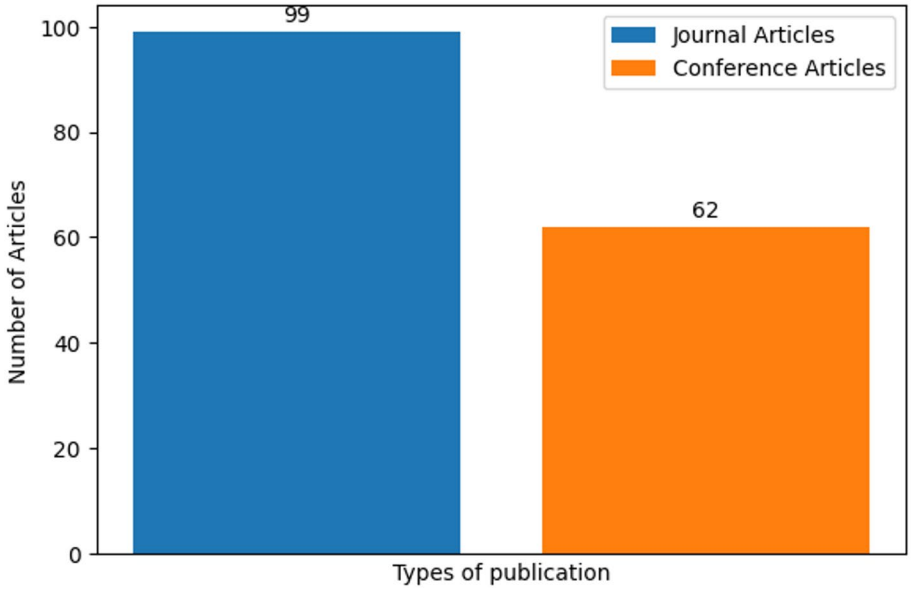


Fig. 7 Categories of articles by publication type

III. *Geographical distribution* Fig. 8 shows the country-wise distribution of the reviewed articles, the data indicates United States holds the highest percentage of publications at 19.88%, followed by India at 16.77% and China at 11.19%. These figures underscore the significant research contributions from regions in North America and Asia. Other prominent contributors include the UK, Netherlands, and Switzerland, indicating a broad international interest and collaboration in IoT and post-quantum cryptography/security.

IV. *Research methodologies used* Fig. 9 shows the types of research methods applied. The data indicates that the majority of articles implemented real-world applications (36.65%), followed closely by theoretical studies (34.78%) and simulation- based

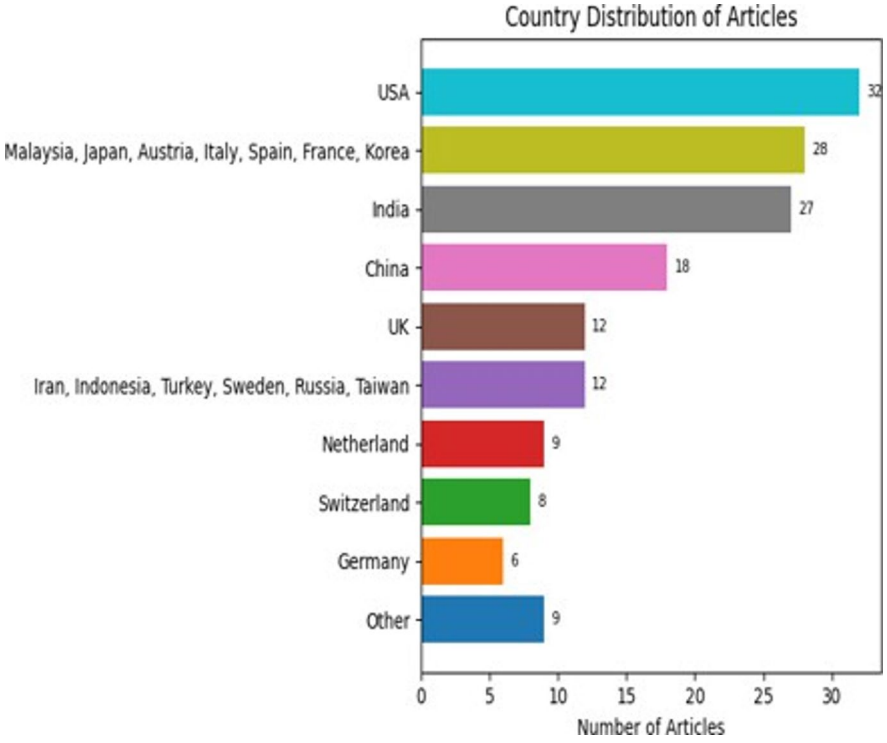


Fig. 8 Countries-based distribution of the reviewed articles

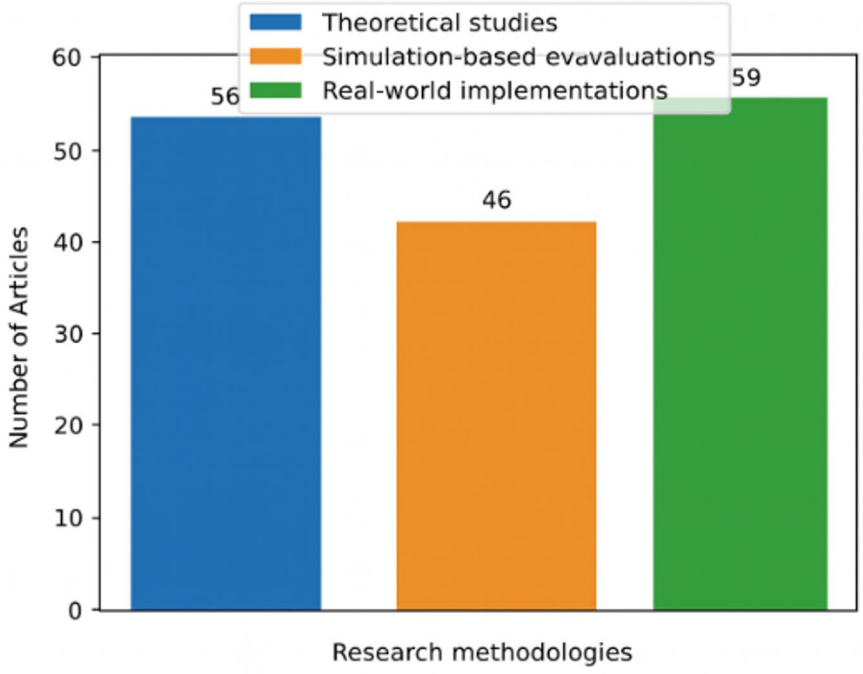


Fig. 9 Distribution of the reviewed papers based on research methodology

evaluations (28.57%). This balance suggests that the research field is conceptually and practically driven, with significant efforts in theoretical innovation and experimental validation.

V. *Target IoT application domains* Fig. 10 shows the IoT application domains covered. The data depicts a significant number of studies (80.75%) falling into the "other" category, indicating they cover various IoT areas. Within the specific domains, Smart Healthcare receives the most attention (4.97%), trailed by Autonomous Vehicles, Industrial IoT, and Wireless Sensor Networks. This distribution highlights a wide range of applications for the research, with a critical focus on developing sectors such as healthcare and transportation.

VI. *Cryptographic approaches* Fig. 11 depicts the cryptographic approaches covered. The data indicates, the most prevalent cryptographic approach is Lattice-based cryptography (28.05%), with a general/unspecified category (51.22%) that encompasses various PQC methods, authentication solutions, and other security measures following closely behind. Hybrid approaches (9.15%) and Quantum Key Distribution (QKD) (4.27%) are also notable in a significant number of studies. This indicates that although Lattice-based techniques are predominant, researchers are investigating a range of secure mechanisms to tackle the challenges posed by quantum-era threats in IoT.

4.2 Thematic analysis

It identifies, analyzes, and reports the recurring qualitative patterns (themes) within the critical reviewed data.

I. *Vulnerabilities in current IoT cryptography and mitigation strategies* Table 5 shows the security vulnerability in IoT cryptography and the mitigation strategies gathered from reviewed papers. The table helps readers to see the problems and how researchers try to fix them.

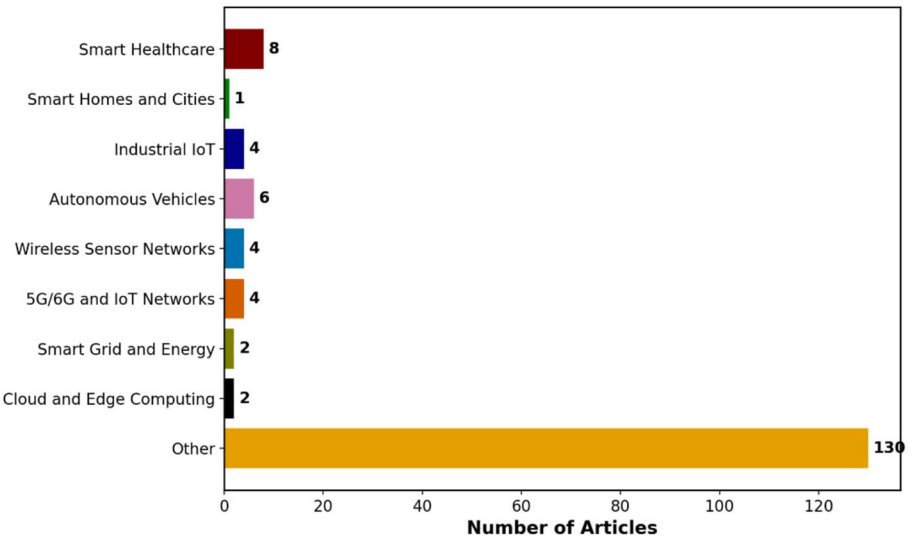


Fig. 10 Reviewed Paper distribution based on IoT application domains

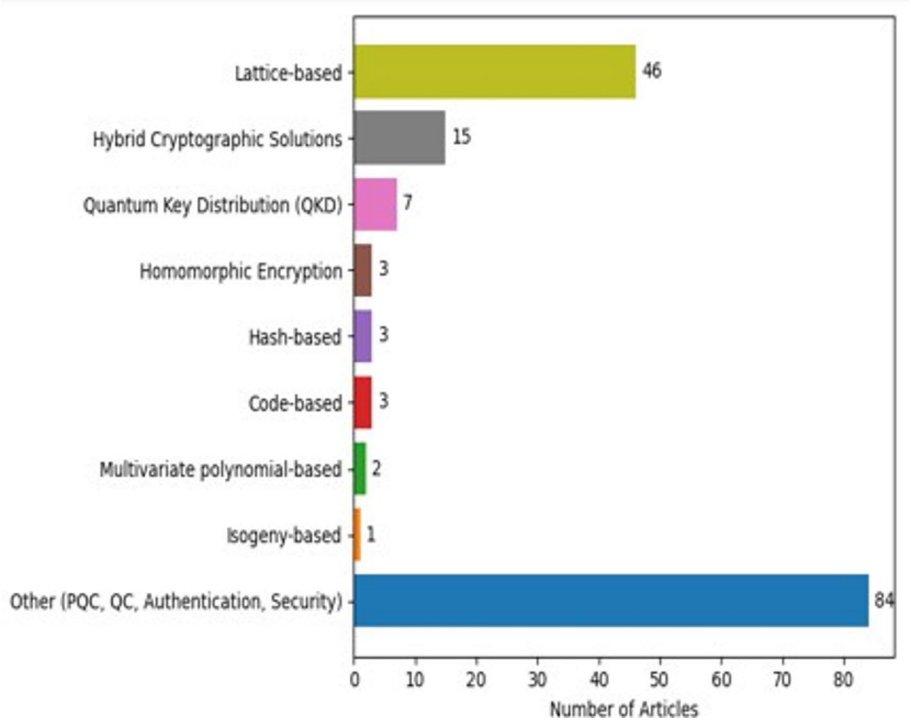


Fig. 11 Number of articles about cryptographic approaches

II. *Suitable PQC algorithms for IoT applications* Table 6 outlines the main features of various PQC algorithms that are appropriate for IoT devices with limited resources, covering aspects such as key sizes, ciphertext dimensions, computational resource demands (such as central processing unit (CPU) and random-access memory (RAM)), energy consumption levels, and their status regarding NIST standardization. Among these, CRYSTALS-Kyber 512 is notable for its minimal resource requirements and readiness for standardization.

III. *Conceptual framework* Fig. 12 discuss a conceptual framework that is structured across three core components: IoT layers, vulnerabilities, and corresponding PQC countermeasures. It illustrates the three foundational IoT layers, namely perception, network, and application that face a range of vulnerabilities and threats landscape [3, 9, 53, 91, 101]. The perception and network layers are predominantly susceptible to implementation flaws, such as side-channel, weak Random Number Generator (RNG), and fault injection attacks; and protocol-level exploits like replay, Key Compromise Impersonation (KCI), and man-in-the-middle (MITM) attacks. These layers currently rely on classical cryptography for key exchange and authentication and making them vulnerable. Critically, the quantum threat from Shor's and Grover's algorithms is a cross-cutting concern, jeopardizing the public-key security cryptographic foundations across all IoT layers. This universal risk necessitates a system-wide cryptographic upgrade.

Table 5 Vulnerabilities in current IoT cryptography and mitigation strategies

Vulnerability	Specific vulnerabilities	Affected cryptographic algorithms	Affected IoT layer	Mitigation strategies
Quantum algorithm attacks	Shor's algorithm (breaks RSA, ECC, and DH); Grover's algorithm (halves symmetric key security)	Public key cryptography, discrete logarithm-based cryptography, symmetric encryption	Network and Application Layer	Adoption of PQC: lattice-based (CRYSTALS-Kyber for KEM, Dilithium for signatures); hybrid RSA/ECC + Kyber for transitional security [61, 63], [66–68].
Resource constraintsand performance issues	Resource constraints of IoT devices (storage, power, bandwidth)	Conventional cryptographic approaches (RSA and public key infrastructure (PKI)	Perception Layer	Lightweight PQC: NTRU and Ring-BinLWE; hybrid ECC + AES for transitional deployment [59, 69], [72–74, 76].
Authentication and identity attacks	Brute Force Attack, Personal privacy leakage, Man-in-the-middle attacks on the quantum channel	Mathematical-based symmetric and asymmetric key cryptographic approaches	Network, Application and Perception Layer	QKD for unconditional security; hybrid QKD + AES-GCM for secure sessions [24, 66], [77–79].
Authentication and identity attacks	Identity privacy compromise during wireless transmission	(Implicitly affects authentication schemes using plaintext identity)	Network Layer, Authentication	Privacy-preserving authentication via PQC-based identity-based encryption (Lattice-IBE) [59, 80].
Authentication and identity attacks	Susceptibility to Key-Compromise Impersonation (KCI) Attacks	Authentication and key exchange mechanisms where compromised secret keys	Authentication, Data Sharing	Multi-layered security with separate PQC-based key generation (Kyber KEM + AES) [24, 59]
Side Channel and physical attacks	Side-Channel Attacks (SCAs), Key Leakage via Fault Injection	BLISS, BLISS-B, SPHINCS+, Dilithium, McEliece	Perception and Network Layer	Side-channel hardened implementations: constant-time arithmetic, masking; Dilithium with hardware resistance [11, 44, 57], [83–85].
Side channel and physical attacks	Vulnerabilities in devices used for quantum key distribution	Implementation of QKD	Perception, Network	Device-independent QKD with privacy amplification & error correction [66].
Network and communication attacks	IoT ransomware, man-in-the-middle (MITM), denial of service (DoS), chosen plain text, chosen cipher text attack	Traditional cryptography	Network layer	Hybrid ECC + Kyber for secure key exchange; lattice-based authentication and session keys [49, 69, 89].
Network and communication attacks	Eavesdropping on classical and quantum communication channels	All cryptographic algorithms re- lying on classical key exchange and Algorithms employing QKD or QC	Network, Application and Perception Layer	QKD with eavesdrop detection; PQC (Kyber KEM) for secure encapsulation [20, 24, 66, 90, 94].
Network and communication attacks	Replay attack	Hash and XOR	Application layer	Nonces & timestamps + PQC-based signatures (SPHINCS+) [65, 96, 97].
Key management and distribution vulnerabilities	Photon number splitting (PNS) at- tacks	BB84	Physical layer	Multi-state qubit QKD (octal SOPs) + hybrid AES-GCM session security [77].
Key management and distribution vulnerabilities	Noise in quantum channels effect on key generation	QKD	Network	Error correction + privacy amplification integrated with PQC KEMs (Kyber) [66].

Table 6 Characteristics and suitability of PQC algorithms for resource-constrained IoT applications [22, 61, 62]

PQC algorithm	Type (KEM/Sig)	Pub Key size (Bytes)	Priv Key size (Bytes)	Ciphertext / sig size (Bytes)	Energy consumption[1]	NIST level	NIST status	Ref
CRYSTALS-Kyber	KEM	800	1568	768	Low (ARM Cortex-M4 benchmarks)	1,3,5	Selected for Standardization (2022)	[62]
NTRU	KEM	930–2401	1234–2983	930–2401	Moderate (mobile/Raspberry Pi tests)	1,3,5	Advanced to 4th Round	[22, 62]
BIKE	KEM	1540–5122	280–580	1572–5154	Moderate–High (Linux/OpenSSL)	1,3,5	Advanced to 4th Round	[62]
SPHINCS+	Signature	32–64	64–128	7856–49,856	High (hash-based operations, memory intensive)	1,3,5	Selected for Standardization (2022)	[62]
Falcon	Signature	897–1793	753–13,953	666–1280	Low (PQClean / x86 results)	1,5	Selected for Standardization (2022)	[22, 62]
CRYSTALS-Dilithium	Signature	1312–2592	2528–4864	2420–4595	Low–Moderate (Raspberry Pi 3/4)	2,3,5	Selected for Standardization (2022)	[61, 62]
HQC	KEM	2249–7245	4044–14,469	4481–14,469	Moderate (embedded tests)	1,3,5	Advanced to 4th Round	[62]
Saber	KEM	992	1248	1088	Low (ARM Cortex-M0)	3	Advanced to 4th Round	[22, 62]
FrodoKEM	KEM	9616–21,520	19,888–43,088	9720–21,632	High (energy heavy on IoT devices)	1,3,5	No longer considered	[22, 62]
Rainbow	Signature	161,600–1,930,600	103,616–1,408,704	66–212	High (impractical for IoT)	1–5	No longer considered	[62]

[1]Energy consumption values are reported qualitatively (Low, Moderate, and High) based on our benchmarking studies across different platforms, where reported results were varied based on hardware/OS: such as ARM Cortex-M4 (IoT devices), Raspberry Pi 4 (Linux), and x86 PCs (PQClean/OpenSSL).

To mitigate these evolving threats, the framework proposes a set of PQC-based countermeasures that are mapped directly to these threats: standardized algorithms, protocol modernization, and secure implementation practices. This includes the adoption of NIST-standardized algorithms like CRYSTALS-Kyber for key establishment and CRYSTALS-Dilithium for signatures to directly counter quantum cryptanalysis. Furthermore, it emphasizes the modernization of communication protocols (e.g., Datagram Transport Layer Security (DTLS)/ Transport Layer Security (TLS) through PQC integration to defend against implementation and protocol-level exploits in addition to the need for secure implementation practices to mitigate side-channel attacks. This holistic approach ensures a resilient transition of establishing a robust quantum-secure IoT ecosystem against modern and future threats.

IV. *Simulation Software and Algorithm* Table 7 depicts a summary of the simulation tools, cryptographic algorithms, and performance metrics utilized in the assessment of quantum-resistant cryptographic protocols for the IoT. The Quantum Cryptography (QC) simulations mainly comprised the execution of QKD protocols like Bennett and Brassard 1984 (BB84), using platforms namely VHSIC Hardware Description Language (VHDL), ModelSim, Vivado, and Java. These simulations were concentrated on measuring communication-related metrics, including bit error rate, encryption duration, and photon interference rate. In contrast, the PQC simulations focused on classical yet quantum-resistant algorithms such as CRYSTALS-Kyber, CRYSTALS-Dilithium, NTRU, Saber, Falcon, and lattice-based techniques. Implementation was carried out using tools such as OpenSSL (Open Secure Sockets Layer), liboqs (lib Open Quantum Safe), Visual Studio Code (VSC), GCC (GNU Compiler Collection), MIRACL (Multiprecision Integer and Rational Arithmetic C Library). Performance metrics measure how well cryptographic protocols work in IoT systems. These include security strength, computation time, memory use, communication costs, and energy use. They help decide if a protocol fits resource-limited devices. Simulation tools use these metrics to test and analyze protocols in virtual settings before deployment. They also guide the design and choice of algorithms, highlighting which perform well under specific IoT limits.

V. *Limitations of PQC standardization for IoT* While PQC is advancing through the NIST standardization process; several limitations remain when considering its deployment in IoT environments. These limitations stem from the mismatch between the design assumptions of PQC schemes and the constraints of IoT devices:

Resource constraints vs. algorithm complexity: Most NIST-selected PQC candidates, such as CRYSTALS-Kyber and Dilithium, were designed for general-purpose computing platforms. Their key sizes, memory requirements, and computational costs are often significantly higher than those of classical ECC or RSA [18, 45, 71, 98, 100]. For IoT devices with only a few kilobytes of RAM and strict energy budgets, this poses significant integration challenges. A signature verification that is acceptable on a laptop may become prohibitive on a battery-powered sensor node.

Communication overhead: Standardized PQC schemes typically introduce larger public keys and ciphertexts [107]. In IoT systems, where communication dominates

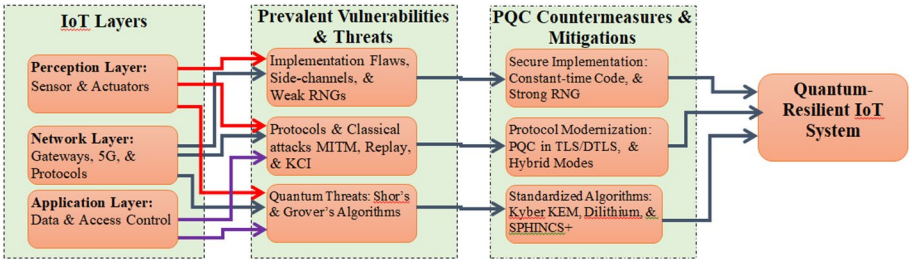


Fig. 12 Conceptual framework for applying PQC to mitigate IoT vulnerabilities

energy consumption, the increase in message size directly impacts battery life and network bandwidth [29, 152]. This is particularly problematic for low-power wide-area networks (LPWANs) such as LoRaWAN or 6LoWPAN, where packet size is already highly constrained.

Lack of hardware support: Current IoT devices have hardware acceleration for AES, SHA, and sometimes ECC, but virtually none provide native support for lattice-based arithmetic or structured polynomial operations [103]. Without hardware assistance, PQC operations must run in software, increasing latency and energy usage. Standardization alone does not solve this gap unless it is followed by corresponding hardware ecosystem development.

Interoperability and legacy systems: Many IoT deployments already operate on long-lived devices with limited ability to update firmware. Standardizing PQC does not automatically guarantee interoperability with existing lightweight security protocols (e.g., Datagram Transport Layer Security (DTLS) for Constrained Application Protocol (CoAP), Transport Layer Security (TLS), and Secure Sockets Layer (SSL) profiles for constrained nodes). Ensuring backward compatibility without adding excessive protocol complexity is still an open issue [59].

Deployment and certification lag: Even after standardization, PQC needs to be incorporated into protocol stacks, operating systems, certification bodies, and compliance frameworks, in which the transition will take years [57]. During this time, IoT systems remain vulnerable to "harvest-now, decrypt-later" attacks, highlighting a gap between theory, standardization, and practical protection.

Table 8 presents the summary of implementation challenges with describing those possible details. It outlines major obstacles in implementing PQC within IoT, primarily resulting from constraints on resources such as significant computational needs, memory limitations, and energy consumption as well as interoperability challenges and adoption barriers including diversity of systems and integration hurdles.

These matters emphasize the attention of research directions with regard to the necessity for PQC solutions that are lightweight, compatible, and efficient in IoT settings as stated under Table 9. This describes the key future directions for PQC in IoT, focusing on optimizing performance, integrating hybrid cryptographic models, and addressing standardization needs for secure and efficient deployment.

Table 7 Summary of simulation tools, algorithms, metrics, and typical IoT layer

Tool	Algorithms	Performance metrics (IoT Priority H/M/L)	Typical IoT layer / workload
VHDL / ModelSim / Vivado / g++	RBLWE-ENC [134]; Kyber / Dilithium / Keccak / Polynomial multiplication [39, 47, 136]	Latency (H); Throughput (M); Signaling/transmission overhead (M); Power consumption (H); Area (LUTs, logic gates) (H); Running frequency (M); SRAM / code size (H)	Device / hardware acceleration / Edge (hardware-offload of crypto)
OpenSSL-1.0.2e, OpenSSL 3.2.1, liboqs	Lattice crypto lib [132]; RSA, ED25519, Falcon, Dilithium; CRYSTALS-Kyber, NTRU, Saber [76, 103, 130, 135]	Execution cost / CPU cycles (H); Communication cost / bandwidth (M); RAM (H); Code size (H); Execution time / latency (H)	Transport / Application (TLS, MQTT gateways, and edge servers)
PQClean / liboqs / C/C++ toolchain (GCC, VSCode, CMake)	Saber KEM, Kyber KEM, Saber-GAKE, compiled Kyber [7]; CRYSTALS-Kyber [135]	Execution time (H); Memory usage (H); Efficiency / clock cycles (H); Computation cost (H)	Gateway / Edge
ContikiOS (firmware for motes)	LLL (Lenstra–Lenstra–Lovász) [127]; GGH / Babai's / Gauss elimination [127]	Latency (H); Execution time (H); RAM footprint (H); Code size (H); Energy (H); Hadamard ratio (M)	Device / Sensor node / 6LoWPAN / RPL
NS-3 + Miracle libraries	Ring-LWE / RLWE experiments [138]	Latency (H); Throughput (H); Signal-to-Noise Ratio (SNR) (M); Packet loss (M)	Network / IoT network simulations (topology, routing, PQC messaging impact)
MATLAB	Rainbow [42]; Multivariate schemes [133]; PQC wavelength multiplexing, BBM92 [139]	Public key size (M); Processing speed (M); Encryption time (M); Confidentiality (qualitative) (M)	physical-layer experiments, prototyping
Vivado + RISC-V toolchain (POClean, riscv32-elf)	Kyber, Dilithium, Keccak; polynomial multiplication, Power2Round, Decompose [39, 47, 136]	Cycle counts (H); Power consumption (H); Area overhead (H); Running frequency (M); SRAM / code size (H)	Device / Edge (constrained processors)
MIRACL (crypto lib)	CRYSTALS-Kyber [135]	Computation cost (H); Execution time (H)	Edge / Gateway
Docker / free5GC / OpenSSL + qcs-provider / Go	CRYSTALS-Kyber, X25519, AES256-GCM-SHA384; classical RSA/DH/ECC; references to Shor/Grover [135]	Latency (H); Bandwidth (M); Throughput (M); Processing latency (H)	Edge network (NFV/5GC experiments)
IBM Qiskit / SimuloQron / CQC / Java platform	BB84 / CD-BB84 [18]; BBM92 [139]; PHOTON-Beetle (hash) [64]	Bit error rate (M); Photon interference (M); Key generation time (L→M for classical IoT, but H for Q-links); #qubits, gate counts, circuit depth (L for now)	Quantum link / future hybrid quantum-classical networks.
Windows 11 + MATLAB 2018	QKD (BB84), OTP, 6D hyperchaotic Chen (6DHCCS), BBM92 [139]	Entropy; Key sensitivity; Computation time; Cyber-attack resilience	backend prototypes
Mosquitto MQTT (broker)	Applying TLS libs like OpenSSL/liboqs	Connection latency (H); Handshake time (H); Payload overhead (M); CPU/RAM on broker (H)	Application layer (MQTT brokers / constrained gateways)

5 Discussion

Due to the widespread use of IoT devices in crucial infrastructures and domains such as healthcare, transportation and smart cities, it is important to protect these devices from different security attacks, including the impending quantum attacks. Protecting IoT devices from quantum-enabled attacks requires examining existing cryptographic vulnerabilities, identifying strategies to mitigate them and the developing quantum-resistant algorithms. This chapter presents the exploration those topics and offers a detailed

Table 8 Existing implementation challenges

Challenge category	Describing the reasons
Resource constraints	High computational cost of core PQC operations (e.g., numeric theoretic transform (NTT), and polynomial multiplication) [29]. The large memory footprint of PQC algorithms and key sizes exceeds IoT device limitations [18, 45, 91], [99–101]. Significant energy consumption associated with PQC processing impacts battery life [54, 55, 102]. Large latency of complex PQC operations and affecting real-time performance [80, 103]. Compatibility problems with older IoT devices lacking the resources for PQC [104, 105]. Challenges in balancing security strength with resource constraints (computation, memory, and energy) [11, 106]. Scalability issues arise from large key and signature sizes in PQC deployments for numerous IoT devices [107]. Difficulty in adapting and implementing complex PQC on diverse and resource-limited IoT hardware [39, 108]. The critical need for lightweight and optimized PQC algorithms tailored for resource-constrained IoT environments [23, 109].
Interoperability	Homomorphic encryption (HE) vs. PQC: Different number systems and dedicated operations [46, 110]. IoT heterogeneity complicates security protocol standardization and deployment [16, 91]. Algorithm integration: Hurdles due to compatibility and software updates [23]. Lattice authentication: Applicable to other systems (e.g., vehicular networks) [102, 111–113]. PQC/QKD deployment in IoT: Challenges in scalability, interoperability, and implementation complexity [14, 114]. PQC interoperability: Major challenge with existing IoT security protocols [9, 46, 59]. Current quantum-resistant authentication doesn't fully address network latency and synchronization [115].
Adoption Barriers	The high cost/complexity of quantum computing limits immediate IoT use [116]. A long PQC transition is challenging, especially for long-lifecycle IoT [57]. Hybrid classical/quantum-safe security models are being developed [100]. Lack of awareness/expertise delays PQC adoption in IoT [117]. Interoperability between PQC and current IoT protocols is a key challenge [59]. Key management and distribution for PQC in IoT is complex [118]. Practical implementation and performance testing of PQC in IoT is lacking [119].

discussion on the impact of quantum computing on IoT cryptographic protocols and the steps necessary to protect these systems against emerging quantum threats.

5.1 Vulnerabilities in existing IoT cryptographic algorithms (RQ1)

The growth in interconnectedness and use of IoT devices in key infrastructures and various sectors has led to an increase in sensitive data exchange which demands stronger security protection [19]. Currently, the security of IoT communication is heavily reliant on classical cryptographic algorithms such as RSA, ECC and DH [2]. These algorithms are based on the computational difficulty of some mathematical problems such as discrete logarithms and integer factorization, for classical computers. However, quantum algorithms that exploit the intensive computational power of quantum computers can efficiently solve these problems [41]. In addition, other vulnerabilities such as algorithmic weaknesses, implementation issues and protocol level vulnerabilities are covered by the existing studies.

I. *Algorithmic Weaknesses* The explored studies underscored that the primary vulnerability of the existing IoT cryp- tographic schemes is an algorithmic weakness that makes them susceptible to quantum computing attacks, specifically Shor's and Grover's algorithms. This weakness is mainly caused by the mathematical problems

Table 9 Future research directions in PQC for IoT

Category	Future research directions
Open research problems	Resource-Efficient PQC: Minimize computation, energy, and memory for IoT [69, 110]. Hardware optimization and co-design: Optimize PQC hardware and explore HW/SW integration for IoT [107, 110, 120, 121]. Lightweight key management: Design efficient key schemes for IoT resource constraints [122–124]. Enhanced lattice crypto for IoT: Tailor lattice algorithms for practical and efficient IoT use [39, 124]. PQC integration and interoperability: Integrate PQC into IoT infrastructure and ensure protocol compatibility [104, 124]. Optimizing specific PQC for IoT: Refine algorithms (e.g., NTRU and BRLWE) for better IoT performance [125]. Improved IoT security with PQC: Enhance security, privacy, and resilience in resource-limited IoT [81, 100].
Hybrid cryptographic models	Hybrid Crypto: Combining classical and quantum security [18, 26, 126]. Cost-effective quantum security (IoT): Hybrid models and PQC [1, 116]. Growing interest IoT: Quantum-safe, hybrid, and lightweight post quantum [1] Hybrid computing: Quantum for complex and classical for simple [17].
Policy and standardization needs	Establishing standards and ensuring compatibility for PQC deployment in IoT [14, 116]. Hybrid crypto for transition: Using PQC combined with existing methods (like ECC) for a gradual shift [91, 95]. Hybrid cryptography: Combine PQC with ECC for transitional security [127]. Prioritized PQC adoption: Focusing on critical IoT sectors like vehicular communication [57]. Adherence to lightweight crypto standards [103]. Balancing resource trade-offs: Addressing traffic with memory usage, especially in constrained IoT [11, 128]. Standardized security benchmarks for PQC in IoT [51, 129].

on which these existing classical algorithms rely, which can be efficiently solved by quantum computers.

(a) *Vulnerabilities in existing Public-Key Cryptography* The majority of existing security protocols used in IoT networks relies on public-key cryptographic algorithms [140]. These algorithms depend on the computational difficulty of certain mathematical problems, which are considered to be unbreakable given the computational capability of classical computers. A common algorithm in IoT networks is RSA. Its security is based on the fact that it is difficult to factor a large composite number into its prime factors. This problem is hard enough for classical computers, but powerful quantum computers are able to factorize these large numbers in polynomial time by using the Shor's algorithm [141]. Using these vulnerability attackers can determine private keys by using the public keys and compromise the whole security of IoT communications.

The other algorithms that are based on the discrete logarithm problems such as DH and ECC are also susceptible to Shor's algorithm. The ECC algorithm is the most widely used encryption algorithm in resource-constrained IoT devices because of its relatively small key sizes compared to RSA for getting equivalent level of security. However, Shor's algorithm can efficiently compute discrete logarithms on elliptic curves, which breaks the security of ECC-enabled systems [142].

(b) *Vulnerabilities in Symmetric Cryptographic Algorithms* For symmetric encryption algorithms such as AES, the Grover's algorithm has posed a significant threat. The

Grover's algorithm, which is a quantum algorithm, can perform an unstructured search quadratically faster than classical algorithms. This means that while it doesn't fundamentally break the underlying mathematical structure of symmetric ciphers, it can significantly reduce the space of the key possibilities that needs to be tried in brute-force attacks. The study in [143] have explained that a symmetric cipher with a 128-bit key would require 2128 operations for a classical brute-force attack but it could potentially be broken with approximately 264 operations using Grover's algorithm. This can be interpreted as the security level of AES-256 against quantum attacks using the Grover's algorithm would be comparable to the security level of AES-128 against classical attacks. Though this does not entirely break the algorithms, it requires the use of larger key sizes to obtain the desired security level in quantum era [140].

Furthermore, Grover's algorithm also impacts the security strength of cryptographic hash functions, which are used to protect data integrity. The authors in [64] have pointed out that Grover's algorithm can be applied to conduct the pre- image attacks on hash functions. Similarly, Grover's algorithm provides a quadratic speedup for such attacks on hash functions and reduces (by a half) the computational operations required to find a pre-image. This effectively halves the security level of the hash functions in the presence of powerful quantum computers. As a result, like the symmetric encryption algorithms, the systems that rely on hash functions needs to double their output size to maintain the same level of security against quantum attacks [64].

(c) *Vulnerabilities in Lattice-based Cryptography* The authors in [144] have conducted a cryptanalysis study on a lattice-based lightweight authenticated key exchange protocol called PiLike [145]. They underscored that the cryptanalysis of this protocol have revealed a critical vulnerability during the private key extraction phase. This security flaw can enable attackers to exploit secret information that obtained during the registration to impersonate legitimate users. The attacker can then negotiate session keys undetected and effectively compromise the protocol's mutual authentication security. The paper also demonstrated that this impersonation attack has a 100% success rate in breaking the claimed security of this PiLike protocol. However, it is important to understand that the identified vulnerability is an algorithmic weakness that is specific to the design of the PiLike protocol, not a general lattice-based cryptographic vulnerability caused by lattice problems.

II. *Implementation Problems* IoT devices are resource-constrained, which are often with limited computational capability, memory storage and energy resources that restrict the implementation of computationally intensive cryptographic algorithms [71]. This constraint often dictates the adoption of simplified or lightweight versions of encryption algorithms, which might have inherent weaknesses or introduce vulnerabilities [146].

(a) *Cryptanalysis attacks* As the security standardization of PQC development is still evolving, several cryptanalysis studies are indicating the weaknesses in some of schemes. For instance, the authors in [144] have conducted a cryptanalysis

study on a lattice-based lightweight authenticated key exchange protocol called PiLike [145]. This asserted that the cryptanalysis of this protocol have revealed a critical vulnerability during the private key extraction phase. This security flaw can enable attackers to exploit secret information that obtained during the registration to impersonate legitimate users. The attacker can then negotiate session keys undetected and effectively compromise the protocol's mutual authentication security. The research also demonstrated that this impersonation attack has more success rate in breaking the claimed security of this PiLike protocol. However, it is important to understand that the identified vulnerability is an algorithmic weakness that is specific to the design of the PiLike protocol, not a general lattice-based cryptographic vulnerability caused by lattice problems.

(d) In addition to this lightweight IoT-specific scheme, some NIST PQC candidates have also collapsed due to different cryptanalysis attacks. For instance, SIKE was considered an attractive candidate especially for constrained IoT devices, -because of its small key sizes and efficiency. However, the authors Castryck and Decru [165] have demonstrated an effective key-recovery attack that exploits torsion point structure and auxiliary endomorphisms in SIDH, which are the basis of SIKE. Their cryptanalytic attack (latest Magma implementation) was able to recover SIKEp434 keys in about 10 min on a single CPU core, which makes SIKE entirely insecure. Similarly, the Rainbow signature scheme, which is based on multivariate quadratic equations, was broken by Ward Beullens [166]. The author presented an efficient key-recovery attack that could extract secret keys of the Rainbow scheme (Level 1) from its public key within an average 53 h (one weekend) of computation time on a standard laptop. These results indicate that even schemes that have been considered promising and advanced by the NIST process can be invalidated by new cryptanalytic attacks and insights.

(e) *Side Channel Attacks* Side channel attacks are a passive type of attacks in which attackers can use information leaked from the physical implementation and execution details of cryptographic algorithms to get the secret keys without the need to target their mathematical foundations and structures [57]. This can involve exploiting different aspects of the devices such as power consumption, electromagnetic emissions, and variations in execution time or emitted frequencies. The distributed and remotely availability of many IoT devices can give adversaries the physical access and time to manipulate the nodes, which increases the risk of such attacks. Several studies [37, 57] have shown that even the modern post- quantum cryptographic schemes can be vulnerable to side channel attacks. Moreover, the study in [147] have discussed that as post-quantum cryptographic schemes evolve, the techniques for analyzing their side-channel resilience will also advance.

Lattice-based cryptographic schemes, which are the well-accepted category of PQC schemes, have been a significant focus of side-channel analysis. For instance, the study in [57] have discussed that BLISS [148] and BLISS-B are lattice-based signature schemes that are vulnerable to practical side-channel attacks using electromagnetic analysis and cache attacks, which can lead to secret key recovery. These attacks have exploited the information leakage related to the norm of the commitment polynomial

and the rejection sampling step. In addition to this, Kyber [38], which is a lattice-based key encapsulation mechanism, has also been targeted by side-channel analysis with researchers proposing message recovery methods targeting the message decoding operation [147].

(d) *Vulnerabilities in key management and distribution* Key management and distribution are the other critical implementation problems. Securely generating, storing and distributing cryptographic keys in a large and distributed IoT devices with resource-limited devices is challenging in terms of storage, computation and communication overhead. Traditional key distribution methods might be inefficient or vulnerable. Public Key infrastructures in asymmetric cryp- tography requires storing only one public and one private key per device, which is helpful with scalability [91]. However, managing these keys across billions of devices, in which many of them have limited updating capabilities remained a challenging task [149].

In response to these problems, promising techniques such as QKD that uses the laws of quantum mechanics to make secure key transmission are gaining great research attention. However, these techniques present their own challenges for large-scale deployments due to their complexity and computational requirement [99]. In addition to this, some technical issues are detected –for example, if an eavesdropper is detected on the quantum channel, the entire process must be terminated and communication cannot resume until the channel is confirmed to be clear [19]. Furthermore, some QKD approaches are vulnerable to attacks such as photon number splitting attacks and faked state attacks [77].

(e) *Fault Injection Attacks* Several studies [41, 57, 147] have also indicated that fault injection attacks are a major threat to the security of IoT devices that should be considered during the implementation of cryptographic algorithms. Fault injection attacks are a type of active attacks where the attackers deliberately insert errors into the device's hardware or software during cryptographic operations. In the case of cryptographic algorithms, the goal of these attacks is to divert the system from its normal execution flow in a way that reveals secret information in communication [57]. Through the observation of how the device behaves under these introduced errors or faults, mainly by analyzing the error outputs, the attacker can obtain the secret information, such as cryptographic keys [57].

III. *Protocol-Level Vulnerabilities* These vulnerabilities exist in the design and the implementation of communication protocols used in the Internet of Things (IoT) and they can be exploited by attackers to compromise the security of the system.

(a) *Vulnerabilities in Authentication protocols* Among the studies included in this systematic review, several [69, 144, 149] of them have explored the authentication protocols used in the IoT networks and discussed their vulnerabilities. The study in [144] have pointed out that some of the existing authentication protocols for IoT fails to achieve mutual authentication, which means that one or both parties

cannot reliably verify the other's identity. This can expose the system vulnerable to man-in-the-middle and impersonation attacks [132]. The lack of sufficient mechanisms to prevent replay attacks in some authentication protocols is also indicated [69]. In replay attacks, the attacker intercepts and retransmits valid authentication messages to gain unauthorized access for a later time [69]. The lack of using timestamps or unique one-time values like nonces in some protocols can create this vulnerability [65].

(f) *Vulnerabilities in Specific IoT Layers* The existing studies have discussed that the vulnerabilities at the protocol level in IoT systems manifest differently across the different IoT layers. The lower layers (physical, perception, sensing layers) rely on low-power communication protocols, such as Wireless Sensor Network (WSN), Radio Frequency Identification (RFID), Zonal Intercommunication Global-standard (ZIGBEE), and Bluetooth. Since these protocols are designed for energy efficiency with low complexity, they can have inherent security weaknesses that expose devices to eavesdropping, where attackers intercept unencrypted signals or exploit weak encryption schemes [19]. Similarly, at the network layer, routing attacks [175], such as sinkhole and wormhole attacks, can compromise the security of the data in transmission. Furthermore, the existing studies have indicated that the application layer protocols which handle sensitive data can be vulnerable to data theft, access control attacks and reprogram attacks [140].

5.2 Mitigation strategies (RQ2)

Several strategies are available in the literature to make the existing IoT cryptographic protocols resistant to quantum threats and fill the aforementioned vulnerabilities. Mainly the following strategies are found in the articles that are explored in this systematic review.

I. *The Adoption of Post-Quantum Cryptography (PQC)* The strategy proposed by most of the studies explored in this systematic review is the replacement of classical algorithms, such as RSA and ECC with PQC algorithms that are believed to be secure against both classical and quantum attacks. These cryptographic algorithms are based on hard mathematical problems that are resistant to the quantum algorithms, unlike the classical encryption algorithms. Depending on the problems they are based on those algorithms can be classified into different groups, such as lattice-based cryptography, hash-based cryptography, code-based cryptography, isogeny-based cryptography and multivariate cryptography [147]. Following the competition launched by NIST, several post-quantum cryptographic algorithms have been developed and submitted for evaluation. As a result, the post-quantum cryptography has attracted a significant research attention, especially the lattice-based cryptography [150]. In addition, this lattice- based cryptography has been frequently stated as the promising solution for IoT security due to their quantum resistance, relatively smaller key sizes and possibility for an efficient implementation [147]. For instance, the study

at [147] have introduced an algorithm called PQ-IoTCrypt, which was built using the Binary Ring Learning with Errors (BRLWE) and optimized for resource-constrained IoT devices. This algorithm uses a symmetric discrete uniform distribution together with efficient polynomial arithmetic to reduce the processing overhead. Moreover, the NTRUEncrypt [65] is another lattice-based cryptosystem that has been explored for constrained IoT devices. However, the remaining challenges are in integrating those PQC schemes into existing protocols (e.g. TLS or SSL), achieving compliance, and ensuring library and hardware support.

II. *Hybrid Encryption Schemes* Several other studies [37, 116] have suggested the idea of hybrid encryption solutions, which combines classical cryptography with quantum-safe solutions to provide both an immediate security and a pathway toward quantum-safe future. For instance, the authors in [37] have introduced a quantum-resistant hybrid encryption for IoT (QRHE-IoT) that combined symmetric encryption for efficiency, asymmetric encryption for secure key exchange and quantum-resistant algorithms such as LWE and Ring-LWE for long-term security. In addition, using QKD for key exchange and PQC algorithms for encryption is suggested as a potential hybrid strategy [65]. Another study in [66] has attempted to use the combination of traditional cryptographic techniques like one-time pad (OTP) with QKD for key distribution. In addition to this, recent study [137] has also focused on benchmarking Kyber and Dilithium across different IoT devices and discussed the trade-offs in execution time, memory usage, and compatibility with existing infrastructure.

However, these studies lack a more actionable transition roadmap that considers the IoT device class (device capacity), the protocol stack used, resource trade-offs and standardization status of the solutions. For example, BLE-based constrained IoT devices can use the hybrid of classical ECC and Kyber-512 KEM (standardized by NIST). This helps to obtain efficiency and interoperability from the classical ECC and quantum resistance from the lightweight Kyber variant (Kyber-512). For Wi-Fi-connected IoT devices with more CPU, memory, and power, a hybrid stack using ECC/RSA for compatibility with existing PKI, with Kyber-768 or Saber KEM, and PQC signature schemes like Falcon or Dilithium, might provide a stronger security without overly sacrificing performance. In more constrained LPWAN or LoRa settings, lightweight PQC signatures such as Falcon can be paired with symmetric keys might be a necessary compromise (because of the message size, energy consumption and bandwidth constraints). At the infrastructure end (edge/cloud/servers), full PQC stacks, such as Kyber for KEM and Dilithium or Falcon for signatures can be adopted for high-impact security tasks. More importantly the standardization and policy dependencies should be central to planning these transition roadmaps.

III. *Quantum Key Distribution (QKD)* The integration of QKD into IoT networks is also suggested to provide a secure way to distribute encryption keys by using the principles of quantum mechanics. It provides the detection of any eavesdropping attempts on the quantum channel [19]. Several research papers have mentioned QKD as a key technology that can be used to secure IoT communication. The BB84 and its variants are mostly discussed in different studies within the context of QKD for

IoT environment. As an example, a flexible continuous-wave quantum cryptography scheme is proposed in [151] for IoT, where an IoT controller can act as a host or client in QKD and efficiently generate quantum random numbers for key distribution and one-time pad communication. Furthermore, a hybrid authentication scheme using QKD for data exchange in the registration phase and classical (though vulnerable) ECC for authentication has been proposed [65]. This clearly indicates the potential for integrating this technique with other methods for quantum resistance.

IV. *Developing Lightweight PQC Schemes* The resource constraints of many IoT devices (limited CPU, power and memory) have forced researchers to focus on the need for lightweight and efficient implementations of PQC algorithms. This includes examining the performance of PQC algorithms in embedded processors and low-cost microcontrollers [152]. The techniques such as streamlined polynomial arithmetic and modified error sampling methods have been also used to reduce computational overhead and enhance efficiency [147].

Several strategies have presented in the existing studies to develop a lightweight PQC algorithm and optimize it for IoT devices. Some of these strategies are for reducing memory usage, storing seeds instead of full private keys for lattice-based schemes, using larger flash memory for storing large public keys [118] and using ciphertext compression techniques [153]. To enhance computational efficiency several strategies are used including, optimized implementations of polynomial arithmetic [54], leveraging assembly language optimizations for target platforms [118] and selecting less computationally intensive algorithms like NTRU [54]. Furthermore, the study in [154] introduced the concept of offloading computationally intensive tasks such as key generation to a more powerful edge computing device.

5.3 Emerging Post-quantum cryptographic algorithms (RQ3)

I. *Lattice-based Cryptography* Lattice-based cryptographic schemes have started to dominate the PQC research due to their strong security proofs and adaptability to low-power devices. Several schemes that are based on Learning with Errors (LWE) and its variants, such as Ring-LWE, Binary Ring-LWE and Module Learning with Errors, have been proposed in different studies [59, 147, 155, 156]. As an example, the PQ-IoTCrypt [147] is a lightweight post- quantum algorithm based on the Binary Ring Learning with Errors (BRLWE) problem, with optimizations for efficient implementation on resource-constrained IoT devices. the study also demonstrated how it reduced computational overhead and resilient against side-channel attacks.

The CRYSTALS-Kyber [38], which is a NIST-standardized Key Encapsulation Mechanism (KEM), have been a focus for IoT applications due to its computational efficiency, which is comparable to some classical algorithms. Its module-lattice structure ensures resistance to quantum attacks with having manageable key sizes. Similarly, CRYSTALS-Dilithium and Falcon were also the focus of several studies for digital signatures. According to the performance evaluations conducted the study in [107] Dilithium2 is one of the best signature algorithms for BLE and WiFi devices in IoT. In addition, NTRU has

been a focus for resource-constrained IoT devices due to its speed and smaller key sizes [104]. Other lattice-based schemes, such as SABER and Ring-LWE based algorithms like FrodoKEM and binary R-LWE, are considered for their efficiency and suitability for resource-constrained IoT devices [157].

II. *Hash-based Signatures* Hash-based signatures are quantum-safe solutions of digital signatures which provides security that relies on cryptographic hash functions. These schemes are considered lightweight post-quantum cryptography mechanisms, which makes them suitable for IoT and other resource-constrained platforms [152]. As an example, the XMSS (eXtended Merkle Signature Scheme) and LMS (Leighton-Micali Scheme) have received NIST recommendations for their use and proposals have been made to integrate SRAM PUFs with the XMSS scheme [155]. Additionally, the SPHINCS+, which is a stateless hash-based digital signature algorithm, has been evaluated within the focus of post- quantum cryptography for IoT devices. Though some assessments have indicated its infeasibility for IoT devices, the integration of this algorithm with TLS on embedded systems have been studied specifically [158]. In general, the role of hash-based signatures in providing quantum-safe security for IoT environments is being actively explored in several studies [119].

III. *Code-based Cryptography* Code-based cryptographic algorithms are built based on the hardness of decoding an unknown error-correcting code and is recognized as resistant to quantum attacks. Among these algorithms, the McEliece scheme is the first code-based algorithm proposed that uses the hardness of decoding a general linear code as a security foundation, which is known to be NP-hard. Its variant, the Classic McEliece scheme was one of the finalists in the third stage of the NIST PQC competition [159]. In the context of IoT environment, the study in [153] has discussed the implementation of the McEliece scheme on a microcontroller. However, the key generation algorithm for the McEliece scheme could not be implemented on the microprocessor due to exceeding the RAM size of the microcontroller. In addition, the hypercube quasi-cyclic (HQC) has gained attention due to its well-considered security metrics in the NIST process, although its computational complexity and memory footprint have traditionally limited its application in IoT devices [107]. Similarly, BIKE (Bit Flipping Key Encapsulation) is mentioned alongside other post-quantum candidate algorithms as suitable for making IoT secure in the quantum era [160].

IV. *Multivariate Cryptography* Multivariate Cryptographic schemes derive their security from the NP-hard problem of solving large multivariate quadratic equations. The Leading candidates are Rainbow and UOV (Unbalanced Oil and Vinegar), which are both NIST Round 2 finalists [152]. Even though there was a concern on the larger key sizes of Rainbow, its new variants specifically optimized to reduce the large key sizes have demonstrated a strong performance on ARM Cortex-M4 platforms. In addition, the HiMQ-3 scheme has advanced this further by targeting 128-bit security on 8-bit microcontrollers [153]. Multivariate cryptographic schemes have significant advantages in quantum computing resistance. However, some mul- tivariate schemes like Rainbow have larger key sizes compared to lattice-based schemes. This can be a disadvantage for implementing them in resource-constrained IoT environments [42].

Recent cryptanalysis breakthroughs have also led to the exclusion of some multivariate schemes such as GeMSS, Rainbow, and Picnic in certain sensitive applications [161].

V. *Isogeny-based Cryptography* Isogeny-based cryptography depends on the difficulty of finding unknown isogenies between elliptic curves [152]. This yields exceptionally small public parameters but they demand intensive algebraic computation [97]. SIKE, a NIST Round 3 alternate candidate for key establishment was considered for IoT application by developing fast hardware architecture [154]. Other studies [97] have indicated that while SIKE offers small parameters which attractive for microcontrollers, it can have very slow runtimes and its security analysis is still at an early stage for signature schemes. The study in [97] has indicated that while SIKE offers small parameters which attractive for microcontrollers, it can have very slow runtimes and its security analysis is still at an early stage with some studies such as [165] demonstrating a successful cryptanalysis attack on it.

In conclusion, lattice-based schemes are indicated as more suitable for IoT devices due to their smaller key sizes, shorter encryption/decryption time, and less memory usage compared to code-based algorithms [45]. On the other hand, the study discussed in [107], the security uncertainties in some lattice-based schemes and indicated that the code-based cryptography as a more deeply understood alternative.

VI. *Case Study Summary - PQC for a Smart Grid AMI* This case study aimed to demonstrate the hypothetical aspect of PQC in providing a necessary path for securing sensitive and long-lived IoT deployments like a smart grid systems against the emerging quantum threat.

- *Scenario* A municipal power company operates an Advanced Metering Infrastructure (AMI) comprising smart meters (perception layer), data concentrators (network layer), and a head-end system (application layer). Smart meters are resource-constrained devices with microcontrollers, limited memory, and strict power budgets that are installed at consumer homes. Data concentrators are gateways placed on utility poles or in neighborhoods that aggregate data from hundreds of smart meters and have more computational resources. Head-end system is located at the central utility office to collects all data, performs billing analytics, and sends control commands (e.g., remote disconnect). Currently system uses classical cryptography (ECC for key exchange and signatures) to secure the communication which makes the system vulnerable to HNDL attacks, where encrypted data is collected for future decryption by a quantum computer.
- *PQC Integration* Company planned a phased based hybrid PQC migration and implementation [130, 178]. A hybrid ECDH/CRYSTALS-Kyber key encapsulation mechanism will be deployed in the TLS/DTLS handshakes to secure the communication of smart meters and concentrators. Then HES commands will be used CRYSTALS-Dilithium for signing critical commands from the head-end system to the grid and ensuring authentication against quantum forgery. Further, SPHINCS + will be adopted for signing long-term integrity sensitive audit logs.

- *Outcome* The migration future-proofed the critical infrastructure against quantum attacks, ensured compliance with anticipated regulations, and improved the overall security posture by incorporating side-channel resistant implementations during the PQC firmware update.

As the smart grid example illustrates, PQC is ready to move from the lab to the real world. Its integration into IoT is a tangible, although demanding, task that requires smart algorithm selections, attention to resource constraints, and a step-by-step migration strategy. The key takeaway is that we can act now to future-proof our critical infrastructure building a defense today that will hold strong against the quantum computers of the future [130, 178].

5.4 Challenges and future directions (RQ4)

I. *Challenges* The following challenges have been identified in the studies included in this systematic review:

- Resource Constraints of IoT Devices: mostly IoT devices have limited computational capability, memory and energy resources, which makes the implementation of complex quantum protocols very challenging [53, 147].
- The standardized and NIST candidate PQC schemes create a higher computational, memory and energy costs. They also have larger key/ciphertext sizes than the classical algorithms, which creates performance, latency, and bandwidth problems for constrained IoT devices [23].
- Integrating PQC schemes with existing protocols: Integrating with protocols such as TLS/SSL is challenging because of compatibility issues, larger key sizes and the lack of adequate libraries and hardware support [116].
- Quantum reversible computing: the quantum-based reversible computing, which is a computational model where every operation is reversible can also cause security problems [19].
- Environmental noise and channel imperfections: quantum systems are highly sensitive to noise and disturbances that comes from the environment. Environmental factors such as fluctuations in temperature, electromagnetic interference and optical losses can lead to errors or security compromise to the quantum communication channels [25, 55, 162].
- Individual attacks: individual attacks on quantum-based communication are possible where an attacker intercepts a quantum signal and creates a new quantum channel [19, 28].
- Managing devices, distributing keys and verifying locations: the integration of position-based quantum cryptogra- phy into IoT environment can create challenges that are related to managing devices, distributing keys and verifying their physical locations [78, 91].
- Cost and size of quantum devices: Quantum devices are currently sizable, unwieldy and expensive, which makes them unaffordable for implement in IoT deployments. The lack of large-scale quantum equipment and devices is a significant challenge [71].

- Limited range of quantum communication: The photons that are used as information carriers in quantum com- munication are limited to the distance over which quantum communication can occur, which makes their use in geographically distributed IoT networks challenging [71].
- Point-to-point limitation: Existing QKD protocols often function only between two devices, which are incongruent with the complex, multi-device nature of real IoT systems that involves thousands of interconnected devices [71].
- Efficiency of qubit communication: Challenges in implementing the physical layer of active optical switches can increase photon losses and curtail the maximum range of the quantum channel that impacts the qubit communication efficiency [19].
- The need for trusted devices and perfect detectors: Some quantum cryptographic tasks such as QKD and quantum random numbers often rely on the assumptions of trusted quantum devices and perfect detectors, which are hard to satisfy in practical environment [163, 167, 169].
- The integration with existing infrastructure: some studies have proposed architectures for IoT applications that consider quantum computing. However, the integration of QKD with the existing IoT infrastructure and protocols without a problem is a challenge [41].
- Scalability, interoperability and implementation complexity: the studies have also indicated that the deployment of QKD in IoT environments may face challenges related to scalability, interoperability and implementation complexity [164].
- Throughput and computational overhead: Some quantum cryptographic solutions might introduce additional through- put and computations on both of the sender and receiver sides, which can impact the performance of resource- constrained IoT devices [140].

II. *Future Directions* The studies included in this systematic review have indicated the following gaps as a potential topic for future research:

- Developing lightweight quantum cryptographic algorithms: in the future researches should focus on designing quantum protocols that are less resource-intensive and can be implemented on the resource constrained devices [100].
- Hybrid Approaches: several studies have suggested focusing on a hybrid security solution that combines the strengths of classical and quantum cryptography for a more practical way to secure IoT in the near future [91].
- Addressing distance limitations: a focus should also be given into quantum repeaters and other technologies to extend the range of secure quantum communication will be important for broader IoT deployments [91].
- Developing robust key management frameworks based on QKD: creating an efficient and scalable key management mechanism for IoT networks that uses the secure key distribution provided by QKD is an important area of research [91].
- Enhancing protection against attacks: future studies should focus on developing quantum protocols and implementations that are more resilient to various types of attacks, including individual attacks and side-channel attacks [19].

- Conducting real experiments and prototype development: conducting experiments and prototype development in real practical IoT environment for quantum cryptographic algorithms is important to validate the theoretical findings and identify the practical challenges [91].
- Quantum Random Number Generators: the use of quantum random number generators to enhance the security of IoT devices and protocols is another researchable area for the future [151].
- Addressing challenges in detection of eavesdropping: the development of a more efficient and less disruptive methods for detecting and handling eavesdropping attempts for IoT is necessary [19].

6 Conclusion

The development of quantum computers is continuously evolving and the existing traditional cryptographic algorithms that are safeguarding IoT devices currently will become vulnerable. As a result, the important services, industries and infrastructures enabled by these devices will also be compromised. As part of addressing this problem, this systematic review was meant to investigate the existing studies that focus on IoT cryptographic protocols within the context of quantum computing. The primary focus of this study was to identify vulnerabilities in existing IoT cryptographic protocols, examine mitigation strategies proposed in the literature, evaluate quantum-resistant algorithms suitable for IoT devices, and outline key challenges along with future research directions. A total of 161 papers were reviewed in this systematic review and the analysis has revealed vulnerabilities caused by algorithmic weaknesses, implementation flaws and protocol-specific issues. More specifically, vulnerabilities of existing public key and symmetric key cryptographic schemes to quantum attacks, particular attacks that exploit the implementations problems such as side channel attacks and fault injection attacks, are identified. In addition, vulnerabilities in key management and distribution, and authentication protocols are also explored by the studies included in this systematic review.

To address these vulnerabilities, existing studies propose several mitigation strategies, including the adoption of post-quantum cryptographic algorithms, the development of classical–quantum hybrid schemes, enhancements to quantum key distribution, and the design of lightweight post-quantum cryptographic solutions tailored for IoT devices. Moreover, challenges in implementing post-quantum cryptography in IoT environments are identified, including resource constraints of IoT devices, environmental noise quantum channels, high cost and limited range of quantum devices, the point-to-point nature of many QKD protocols, and complexities in scalability and integration with the existing infrastructure. To address the gaps, future studies should focus on developing lightweight and hybrid quantum cryptographic protocols, enhancing key management and distribution, addressing quantum distance limitation and enhancing protection against specific security attacks. The significance of this systematic review lies in providing a theoretical framework for anticipating and mitigating future security threats on the existing and trending cryptographic algorithms. Furthermore, this systematic literature review discusses the key challenges in IoT cryptography under emerging quantum threats and highlights promising directions for future research. This will provide a

valuable guidance for researchers, academicians and industry practitioners in strengthening the resilience of IoT systems against quantum-enabled attacks.

Supplementary Information

The online version contains supplementary material available at https://doi.org/10.1007/s43926-025-00275-6.

Supplementary Material 1.

Acknowledgements
The authors would like to take this opportunity to acknowledge Ethiopia Information Network Security Administrations and Adama Science and Technology University for providing the necessary facilities to support this investigation.

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Ketema Adere : Conceptualization, Methodology, Supervision, and Research execution Seble Hailu : Conceptualization, Conducting surveys, Resources, and Project administration. Alemybrah Tseghai : Methodology, Formal analysis, and Manuscript preparation.Gizachew Haile : Investigation, Surveys, and Project administration.Bereket Tesfaye : Investigation, Visualization and Manuscript preparation.Wudu Bitew : Conducting surveys, formal analysis, and manuscript preparation. Mesfin Abebe : Research methodology, Programs verification, Review, and Editing. Ararso Kemal : Conducting surveys, Formal analysis, Review, and Editing.Semere W/Silassie : Article data collection, Validation, Visualization, and Editing. Dereje Regassa : Mathematical analysis, Research investigation, and Visualization. Zelalem Gezachew : Conducting surveys, Compiling, and Managing articles data. Wengelawit Alemu : Article Data Collection, Management, and Visualization.

Funding
No funding was available for this work.

Data availability
No datasets were generated or analysed during the current study.

Declarations

Ethics approval and consent to participati
Not applicable.

Consent for publication
Not applicable.

Competing interests
The authors declare no competing interests.

Received: 1 August 2025 / Accepted: 26 December 2025
Published online: 27 January 2026

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